Faithful attribution recovers the causal wiring, not the semantics: a ground-truth audit of interpretability on a differentiable microprocessor

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Abstract

Interpretability methods are judged by their faithfulness: whether they name the true causes of a system's output. The field treats faithfulness as the test that separates a real explanation from a merely plausible one. We test this on the Understanding VCS Project, a bit-exact and differentiable copy of the Atari 2600. The machine was never built to be interpretable. Yet it is fully known, so the true cause of any output can be computed by exact intervention. We score thirty interpretability methods against this known answer. Causal and intervention methods are more faithful than gradient and correlational ones (**0.66** versus **0.37**; the gap excludes zero). The clearest case is a sprite's position, where the naive gradient is provably zero: a causal method scores **1.000** and standard saliency **0.165**. But faithfulness is not what is missing. When we switch on the port's differentiable sampler, the naive gradient becomes non-zero—saliency rises from **0.000** to **0.791** on Pong—yet its faithfulness stays low on most games and no method recovers what a variable means. Every method reports which signals matter, never what they mean. We name this remaining distance the semantic gap, and we explore it in detail. Meaning is not in the mechanism. For software it is fixed outside the code, in the documentation and the data the program was written for. No method that reads only the mechanism can reach it. The Understanding VCS Project offers a reusable benchmark with known answers, for explainable AI, mechanistic interpretability, and software verification.

Keywords: interpretability, explainable AI, mechanistic interpretability, faithfulness, ground truth, causal attribution, semantic gap, Atari VCS

Interpretability now matters in practice. Neural networks drive cars, read scans, and approve loans. Before we trust such a decision, we ask why. Yet on the systems we most want to explain we cannot check whether the answer is correct. A saliency map for a deep network can look convincing and can change with the action, and still not be a correct explanation [1]. The field of explainable AI has converged on *faithfulness* as its quality criterion: whether an explanation names a system's *true* causes [2]. The difficulty is that on a real neural network there is nothing to measure it against. Sanity checks can show that a method fails, but with no reference map they cannot certify that it is correct [3].

This blind spot is not specific to deep learning. Jonas and Kording applied standard neuroscience methods to a microprocessor in which every transistor is known [4]. Lesions, tuning curves, and connectomics produced rich structure, but none of it recovered how the chip computes. The same gap recurs across the three interpretability traditions that rarely cite one another. The neuroscience battery records and perturbs a system. The attribution tradition asks which inputs mattered for a decision, through gradient and saliency maps [5–8], class-activation methods [9, 10], occlusion [11–13], and surrogate or game-theoretic attributions [14, 15], applied to Atari agents to visualise what a policy attends [1, 16]. Mechanistic interpretability aims past attribution at the circuit, through path patching and causal mediation [17–20], alignment search and interchange interventions [21, 22], and the sparse dictionaries and probes that read features off the state [23–27]. Across all three, methods are validated against each other or against human intuition rather than against the mechanism [28], and even the sparse-dictionary branch reports that its tuning proxies do not track interpretability [29]. Faithfulness has stood in for understanding, and no one could test the stand-in.

Maier et al. recently built the missing reference. It is a bit-exact, differentiable port of the Atari 2600 Video Computer System (VCS) that reproduces a real 1977 machine to the byte and pixel [30, 31]. The VCS is software with a known specification, so the true cause of any output is computable by exact intervention. This *intervention oracle* (Section 3.2) returns the true causal effect for any candidate cause of any output, measured rather than estimated. It operates over the platform shown in Fig. 1. On this machine an interpretability claim can be scored right or wrong. We operationalise interpretability as ground-truth recovery: an explanation is interpretable to the degree that its claimed structure matches the system's true causal structure, scored by the oracle.

We audit 30 interpretability methods, plus the oracle as a positive control, on two axes: faithfulness against the oracle, and a documented plausibility proxy for how convincing a method looks. The first result is the expected one, and we treat it as a step rather than the finding: causal and intervention methods are faithful, while gradient and correlational methods are not, most sharply on the discrete sprite-position outputs whose naive gradient is provably zero. A reader could dismiss that zero as an artifact of a discrete index, so we remove it: turning the differentiable port's bilinear sampler on restores the position gradient, and the gradient methods become faithful too on the games with a moving tracked sprite. Making them faithful does not change the conclusion, because faithfulness was never the dividing line.

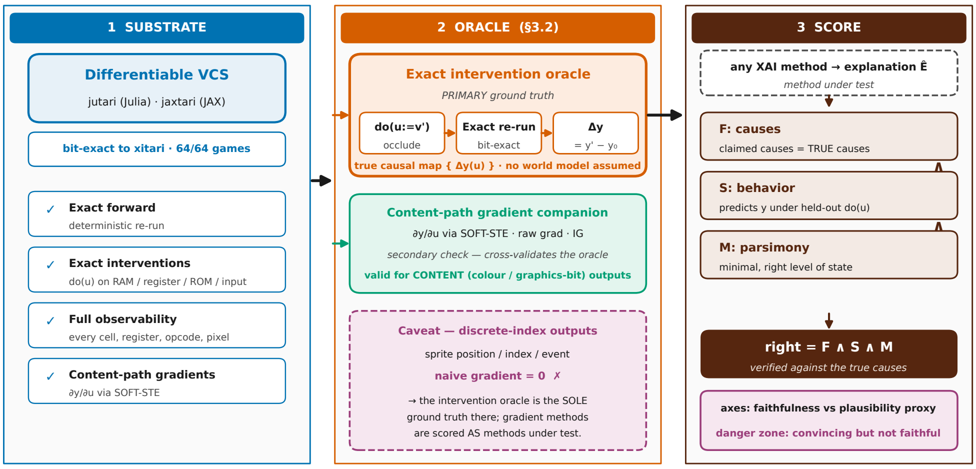

Fig. 1 **The measurement platform and its ground-truth oracle.** A differentiable, bit-exact port of the Atari VCS (left), realized as `jutari` (Julia) · `jaxtari` (JAX), gives four capabilities no neural network grants for free: exact forward, exact interventions do(u), full state observability, and content-path gradients. On it we build the oracle (centre): the primary intervention oracle of Section 3.2, do($u{:=}v'$) $\to \Delta y(u)$ by bit-exact re-run, with a content-path gradient $\partial y/\partial u$ as a companion on the content path only. Any explanation $\hat{E}$ is scored by the correctness triad (right), right = F $\wedge$ S $\wedge$ M, and reported on two axes, faithfulness against the oracle versus a plausibility proxy. Faithfulness aggregates correlation, sign and magnitude agreement, deletion/insertion area-under-curve, and precision at k against the true causal top-k (Section 3.4). A discrete index — a sprite's position, set by strobe timing — has a naive gradient that is zero under the hard forward map (Prop. 1); for position, index, and event outputs the intervention oracle is the *sole* ground truth and gradient methods are scored as methods under test. The schematic draws no experiment data. Data and oracle: see Supplement; oracle defined in Section 3.2.

With the same oracle we ask what a faithful method *delivers*, and the answer is always the same kind of object: a causal-effect map or a data-flow graph, never the variable's meaning or the routine's algorithm. We call an explanation right only if it is **F**aithful (its claimed causes are the true ones), **S**ufficient (it regenerates behavior under held-out interventions), and **M**inimal at the algorithmic level (right = $\mathbf{F} \wedge \mathbf{S} \wedge \mathbf{M}$, Section 3.4). The data then show a gap in both directions: a circuit that is graph-correct and minimal yet does not reproduce the behavior it explains, and a learned feature that matches a named game variable perfectly yet is causally inert. We name this residual the *semantic gap*, and it does not shrink when faithfulness is made universal. The per-method scores are reported in Section 1 (Tables 1–3 and Figs. 2 and 3).

The reference for meaning lies outside the mechanism. Every semantic label in the study entered from outside the system, imported from external annotation sets [32, 33] and only *causally verified* on our substrate, never *discovered* by a method under test. An alignment method that reports perfect accuracy is therefore right by construction, because it checks a variable we supplied. Probing shows that information is present, not that it is used [25], and our oracle is the strictly stronger test. What fixes the meaning is the system's *behavior*. For software, that behavior is set down in its documentation and data: the manuals, the specification, and the design. IEEE 610.12 makes these part of the definition of software itself [34]. The wiring is in the mechanism, so any faithful method can recover it. The meaning is not, and no method we tested supplied it.

To our knowledge this is the first cross-tradition audit of interpretability methods on a real, human-written software artifact whose ground truth is independent of it, scored by an exact intervention oracle; we claim no unqualified "first," since synthetic known-answer benchmarks and Jonas and Kording's microprocessor study [4] precede us. We read a 1977 chip as a necessary-condition screen, not a predictor of success on learned networks: under perfect ground truth and no learning, failure here is strong negative evidence for harder, less observable systems, and the chip still shows the properties that make interpretation hard, from aliased RAM cells to race-the-beam timing (Section 2). We are explicit about the limits: no method recovered the program's meaning, algorithm, or design from scratch, and the plausibility axis is a documented proxy, not a human study. This paper *defines and measures* the semantic gap and locates its reference in the system's behavior; *closing* it is future work, hopefully inspired by this paper. The hard part of interpretability is not faithfulness but recovering meaning, which remains unsolved even when faithfulness is achieved. We believe that naming and demonstrating the existence of this gap is a first step toward closing it.

1 Results

1.1 The neuroscience battery

We begin with a known concern in the field. Jonas and Kording asked whether a neuroscientist with the standard toolkit could understand a microprocessor. They found that the toolkit yields rich structure but misses the mechanism [4]. They could show the failure but not measure it. On the 6502 they had no causal ground truth to score each method against. We supply that missing number. We run the same battery on the bit-exact, fully known VCS, where exact intervention recovers the true causal role of every register and RAM cell (Section 3). This makes Kording's qualitative lesson quantitative. Across the eight analyses A1–A8, faithfulness against the oracle ranges from 0.05 to 1.0 and averages 0.45 [35]. The oracle-as-method positive control sits at 1.0 by construction (Table 1). The structure these methods report is real. But structure is not the same as cause. The score measures the difference.

The dependency structure is largely recoverable. A1 reconstructs the dependency graph over state variables, a connectome of read/write edges, and scores it against the true data-flow graph at a faithfulness of 0.34 [35]: most of its edges are spurious. The single-unit lesion A2 fares best. It clamps each RAM cell in turn and ranks it by the behavioral damage it does. The resulting importance map has a Spearman correlation of 0.99 with the unit's true causal role (mean 0.992 over the six core games) [35]. Lesioning is causal, which is why A2 is the one classical method that scores near the top.

Yet a near-perfect rank order still misses interactions. A single-unit lesion changes one cell at a time, so it is blind to causes that act only together. On Space Invaders the per-unit ranking misses 40% of the interactions the oracle records. One super-additive pair swings behavior by up to 1044 screen-pixels beyond the sum of its parts [35]. The ranking of which units matter is correct, but the model of how they combine is missing. So even the strongest classical method recovers a ranking, not the computation. An effect map does not show the algorithm behind it.

Tuning curves report structure that does not match the true cause. A3 fits per-unit tuning to a game variable, the workhorse of systems neuroscience. On the VCS it actively misleads. Its *spurious-tuning rate* is the fraction of strongly-tuned units whose tuning does not match their true causal role. That rate is 0.50 on Pong, and its faithfulness averages 0.09, the lowest in the battery [35]. Tuning is correlational. On a machine where many cells co-vary with the beam clock, apparent tuning is cheap. This is the same present-versus-used distinction we discuss later for linear probing.

Correlation structure and field potentials are real, but they are mostly side effects, not causes. A4, the spike-word analysis of pairwise and global correlation, reproduces the familiar weak-pairwise/strong-global signature. Yet against the true coupling its faithfulness is only 0.22, with sufficiency near zero ($S = 0.026$) [35]. A5 is the local-field-potential analogue that pools regional activity into power spectra. It scores 0.30, and much of that variance is the known frame and scanline clocks [35]. Both methods detect the machine's activity but not its computation.

Granger causality treats "happens earlier" as "is the cause." A6 infers directed CPU/TIA/RIOT edges and scores them against the true data-flow. On Pong its faithfulness is 0.0, with false-edge and missed-edge rates both at 1.0 [35]. The VCS is clock-locked, so nearly everything is predictable from everything else one cycle earlier. Equating precedence with causation therefore infers spurious edges everywhere. Neither a longer lag nor a stricter threshold fixes this. Precedence and causation differ here, and only the intervention oracle separates them.

Dimensionality reduction scores in the middle, with NMF and PCA tied. A7 decomposes the state tensor into latent components and matches them to the known clock, read/write, and vsync signals. Non-negative matrix factorization and PCA both match about 60 percent of the known components on average (at best 80 percent on a single game), so neither factorization is the more faithful at this analogue [35]. NMF leads only on the secondary recon-error/FSM composite, where its non-negativity better fits the additive register basis. This makes A7 the second-highest score, behind only the causal lesion. It still leaves 40 percent of the components unmatched.

A complete record is still not an explanation. A8 records the full RAM and register state, faithful and sufficient by definition ($F = 1.0$, $S = 1.0$). Yet its minimality is only 0.08 [35]. At the algorithmic level, a recording of every cell explains nothing. Faithfulness and sufficiency without minimality give a log file, not an account of the computation. This is the first sign that the three axes are independent. The semantic-gap result makes that point central later.

Taken together, the battery confirms the expected result and puts a number on it. The advance over Jonas and Kording is that we grade the *architectural* register-transfer level with a *quantitative* F against an exact oracle, rather than giving a qualitative verdict on the transistor. The contrast is clear, because the causal operators we apply to the same machine in Section 1.2 and Section 1.3 reach the ceiling this

Table 1 The neuroscience battery scored against the intervention oracle. Faithfulness F of each classical analysis A1–A8 against the exact intervention oracle (Section 3.2), with its 95% bootstrap confidence interval over the six core games and the documented plausibility proxy (Section 3). The battery averages 0.45; the oracle-as-method positive control is 1.0 by construction. Only the causal single-unit lesion (A2) reaches the ceiling on a non-trivial metric. The whole-state recording (A8) is at 1.0 only because recording everything is faithful, yet its minimality is 0.07 — a log file, not an account. Every value reads from a committed per-game record via the leaderboard [35]; none is re-run or estimated.

Analysis	F (95% CI)	Plaus	What F scores
A1 connectomics	0.34 $[0.09, 0.67]$	0.55	data-flow-graph F_1 vs. oracle
A2 single-unit lesions	0.99 $[0.99, 1.00]$	0.55	lesion-importance ρ vs. true role
A3 tuning curves	0.09 $[0.00, 0.25]$	0.80	1−spurious-tuning rate
A4 spike-word correlations	0.22 $[0.07, 0.41]$	0.80	coupling vs. true coupling
A5 local field potentials	0.30 $[0.22, 0.39]$	0.80	1−clock-explained variance
A6 Granger causality	0.05 $[0.00, 0.11]$	0.80	1−false-edge rate vs. data-flow
A7 dimensionality reduction	0.60 $[0.60, 0.60]$	0.75	NMF/PCA matched-component fraction
A8 whole-state recording	1.00 $[1.00, 1.00]$	0.30	complete record (minimality $M = 0.08$)
Oracle (positive control)	1.00 $[1.00, 1.00]$	1.00	exact intervention ground truth

battery cannot. So the limitation lies in the analysis methods, not in the machine. We mean this strictly as calibration. Even the battery's best members recover a ranking, a graph, or a record, never the algorithm the program runs. This remaining gap is what the paper measures.

1.2 Attribution: which inputs caused the output

Attribution returns a map of which inputs mattered. We apply these methods to the VCS to ask a question a real network cannot answer: is the map true? Every attribution is scored against the same intervention oracle (Section 3.2) that defines ground truth. For each chosen output y, a pixel, a score, or a game event, we know the exact set of causes it should have named. We ran twelve attribution methods across six core games (Pong, Breakout, Space Invaders, Seaquest, Ms. Pac-Man, Q*bert) and an oracle-as-method positive control, and score each against the intervention oracle in Table 2. We score all methods from a shared live state rather than a boot frame. We seed a single random-action prefix of 90 steps and a horizon of 15 (seed 0), gate the state through the oracle's cause-density check so that degenerate frames are rejected, and read every method from the same screen-buffer region (Section 3). The gate matters: on Seaquest it lifts the count of live causes from 2 to 19 of 48 candidates, so the methods are scored where the program is actually computing.

The main result is a split between two output types, not a ranking of methods. A content output is the value of a register, a colour, or a graphics bit. It is a differentiable function of the state, so its gradient flows. A position output is where a sprite is drawn, set by strobe timing on a discrete index. It has provably zero naive gradient (Paper 1, and Section 3). Hence the whole gradient family hits this limit, and it splits the leaderboard into two clear groups.

On the content path the gradient is faithful, as expected. Vanilla saliency [5] puts most of its attribution on the true causal byte: on Q*bert its attribution of a content read correlates with the oracle at Pearson 0.999 with precision@3 of 1.0, on

Ms. Pac-Man at 0.811 / 1.0, on Breakout at 0.750 / 1.0. Integrated Gradients [6], Grad×Input/DeepLIFT [36], SmoothGrad [7], and guided backpropagation [8] all reproduce this, because the output we explain is a one-hot read whose Jacobian $\partial y/\partial u$ is a constant pointing at the read index. Smoothing does not change the score ($\Delta\,\mathrm{corr} \approx 0$), and guided backprop reduces to vanilla saliency. On this path all methods give the same answer.

The gradient family is weak on the position output. The naive gradient there is identically zero, so the raw saliency, Grad×Input, SmoothGrad, guided-backprop, and Integrated-Gradients maps carry no signal through the position read (Prop. 1). A differentiable substrate can restore a non-zero gradient here, and we run that repair in Section 1.4, but the restored gradient is faithful only on a minority of games and its faithfulness stays low. Averaged over the position regime the gradient and correlational family reaches only 0.248 against the oracle, against 0.486 for the causal and intervention family (Table 2). Across all output regimes the gradient and correlational family scores 0.367 and the causal and intervention family 0.664. The position map is not empty by choice; it is weak exactly where the discrete game logic sits.

Two further results make this point stronger. The baseline choice is the perennial worry about Integrated Gradients. Here it changes magnitude, not faithfulness. The $(x - x_0)$ factor moves $\max|\mathrm{IG}|$, yet the ranking and the oracle correlation hold, because the constant one-hot Jacobian keeps the mass on the same byte. Only the degenerate baseline equal to the content byte breaks it. Expected Gradients [37] averages IG over a distribution of recorded VCS states. It removes that degeneracy and is provably stable across reference draws (mean pairwise cosine ≈ 1). Yet, on the content path it scores the same faithfulness as a well-chosen single-baseline IG at $M\times$ the compute, and it still vanishes on position. The more careful method gives stability but not accuracy. Its all-regime faithfulness of 0.062 is the lowest in the study.

Guided backprop and vanilla saliency fail the program-randomization sanity check of Adebayo et al. [3] on the system itself. We scramble the ROM so the program boots to a completely different state ($\sim$69–127 of 128 RAM bytes change, lowest on Seaquest), and the content-path saliency map stays byte-identical (cosine ≈ 1.0 on all six games). The Jacobian $\partial y/\partial u = e_{\mathrm{idx}}$ is a constant one-hot, independent of what the program computes. So the map points at the read index no matter which program produced the output. The saliency is therefore provably model-invariant. A faithful explanation must change when the model changes. This one does not. The intervention oracle, by contrast, distinguishes the two programs at once. The map identifies the right cause but does not depend on the model that produced it.

When we replace the gradient with a real intervention, the position result improves. Occlusion [11] sets a candidate cause and re-runs the bit-exact program, a coarse version of the oracle. On the position regime, where the naive gradient is zero, it reaches faithfulness 0.568, and its all-regime faithfulness of 0.639 is the highest among the attribution methods. The black-box perturb-and-re-run methods follow the same logic and result: KernelSHAP / Shapley sampling [15, 38] (0.645), LIME [14] (0.627), and RISE [13] (0.548) each draw masked coalitions and average over the genuine responses, so each works on position outputs where the gradient is dead, and each reports its own completeness ($\Sigma\phi = y(\text{intact}) - y(\text{all-masked})$ for KernelSHAP, the surrogate R^2 for LIME).

The two minimal-mask methods meet Atrey et al.'s [28] objection that perturbations stray off-manifold. Extremal perturbation [12, 39] optimises an area-bounded mask of real *do*(occlude) re-runs, and the on-distribution counterfactual [40] substitutes candidate cells toward a genuine alternative VCS state, the RAM of another frame of the same ROM. Both stay on the data manifold by construction and both work on position outputs (position-regime faithfulness 0.278 and 0.098, all-regime 0.401 and 0.217). The counterfactual is a no-op on three games, where no on-distribution content byte varies between frames, so its map is flat and we mark those cells invalid rather than score them. The cross-tradition contrast is clearest across all three phases: the causal and intervention family averages 0.664 faithfulness [95% CI 0.60–0.74, $n = 11$] against the oracle, against the gradient and correlational family's 0.367 [0.32–0.41, $n = 14$] (Table 2). Restricted to the attribution methods alone the two means are close, 0.419 for the three intervention methods against 0.443 for the gradient methods, because the content-path gradient is faithful and lifts the gradient mean. The intervention advantage is a cross-tradition and position-regime effect, not an attribution-wide one, and we report it as such. A perturbation is faithful only when it is a valid intervention.

Some popular methods return no answer at all, because the VCS has no structure for them to use. Grad-CAM and Grad-CAM++ [9, 10] need a final convolutional layer to weight; attention rollout [41] needs attention matrices; VIPER [42] needs a learned policy to distil. The VCS is a 6502 CPU with a TIA and a RIOT executing a fixed ROM. Across all six games the count of each required substrate is exactly zero, against ≥ 16 genuine candidate causes per game for the applicable methods. We record this as a measured finding, `applies` = `false`, not a poor score: the absence is structural and the same in every game, and re-targeting Grad-CAM's pooling onto raw register state would only be the content-path gradient renamed. Several well-known methods produce no result on a real artifact. This absence is itself a finding of the audit.

Across the twelve methods, faithfulness and plausibility move in opposite directions: methods rank high on plausibility but low on faithfulness. The clearest pair makes it concrete. On the position regime the causal method activation patching (Section 1.3) reaches faithfulness 1.000, the oracle ceiling, on all six games, while vanilla saliency, a widely used baseline, reaches only 0.165. This is a per-method position gap of 0.835. Yet, on the plausibility proxy of Section 3 the popular method has the higher score (0.9 versus 0.5): the weaker map has the higher plausibility score. Low faithfulness with high plausibility is the danger zone. On the VCS we can measure it directly, not just warn about it.

We claim only that these maps are faithful, nothing more. A correct occlusion map names the true causal bytes. It does not tell us what those bytes mean, that a cell is the ball's x or that a routine is the bounce. Object-level semantics enter only through the external T3 labels (AtariARI, OCAtari; Section 3), and even there we verify a supplied label by intervention rather than discover it. Whether a faithful attribution amounts to understanding the computation we take up in Section 1.4. Here we have established only that faithfulness is achievable, and that the most popular methods do not achieve it.

Table 2 Attribution scored against the intervention oracle (Section 3.2). Twelve attribution methods on the six core games (oracle-as-method positive control = 1.0). F is the all-regime faithfulness against ground truth with its 95% bootstrap CI; F_{pos} is the position/index regime alone, where the naive gradient is provably zero; *Plaus* is the documented method-tradition plausibility *proxy* (Section 3), *not* a human measurement. The naive gradient is provably zero on the position read (Prop. 1); the small non-zero F_{pos} the gradient methods still carry comes from the content channel and, for the two SmoothGrad-family rows, from the sampler-restored gradient of Section 1.4, which is faithful on only a minority of games and stays low. Across all regimes the causal and intervention family leads (0.66 against 0.37, position 0.49 against 0.25), and the popular gradient methods carry the *higher* plausibility tag — the danger zone, measured. The two family-mean rows are the cross-tradition families ($n = 14$ gradient/correlational, $n = 11$ causal/intervention across all three traditions), not the twelve rows above; restricted to attribution alone the two means are close, 0.44 gradient against 0.42 intervention, because the faithful content gradient lifts the gradient mean. The on-distribution counterfactual is a no-op on Space Invaders, Ms. Pac-Man, and Q*bert (no on-distribution content byte varies between frames, so $\Delta = 0$ and the cells are marked invalid rather than scored). Grad-CAM, Grad-CAM++, attention rollout, and VIPER [9, 41, 42] do not apply (no convolution, attention, or learned policy) and are omitted. Values read from the committed leaderboard [35].

Method	Family	F (95% CI)	F_{pos}	Plaus
KernelSHAP [15]	black-box	0.65 [0.51, 0.77]	0.58	0.90
Occlusion [11]	interv.	0.64 [0.50, 0.76]	0.57	0.55
LIME [14]	black-box	0.63 [0.50, 0.76]	0.55	0.90
RISE [13]	black-box	0.55 [0.43, 0.69]	0.42	0.90
Grad×Input / DeepLIFT [36]	gradient	0.52 [0.43, 0.62]	0.16	0.90
SmoothGrad [7]	gradient	0.42 [0.37, 0.46]	0.16	0.90
Vanilla saliency [5]	gradient	0.42 [0.38, 0.46]	0.16	0.90
Integrated Gradients [6]	gradient	0.42 [0.37, 0.47]	0.14	0.90
Extremal perturbation [39]	interv.	0.40 [0.28, 0.51]	0.28	0.55
Guided backprop [8]	gradient	0.34 [0.19, 0.46]	0.00	0.90
On-distribution counterfactual [40]	interv.	0.22 [0.05, 0.43]	0.10	0.55
Expected Gradients [37]	gradient	0.06 [0.00, 0.19]	0.05	0.90
Gradient/corr. mean		0.37 [0.32, 0.41]	0.25	—
Causal/interv. mean		0.66 [0.60, 0.74]	0.49	—

1.3 Mechanistic interpretability on a known circuit

Mechanistic interpretability makes the strongest claim in the field. It tries to explain how a system computes, not just where it attends. Its methods read internal activations as a circuit and try to recover that circuit. On a neural network the recovered circuit can only be checked against itself, because the true wiring is unknown. The VCS removes that limitation. Its data-flow is the program's data-flow, known exactly from the disassembly. Every intermediate value is a RAM cell or a register we can read, clamp, and re-run bit-exactly. We therefore run ten mechanistic-interpretability methods on a circuit whose answer we already know. We score each against the same Section 3.2 oracle: the exact intervention table, the true read/write graph, and the $\mathrm{F} \wedge \mathrm{S} \wedge \mathrm{M}$ triad of Section 3. This calibrates the toolkit against a *known* circuit inside a real, human-written artifact rather than a synthetic or compiled one. Table 3 places the mechanistic methods against the oracle. The result splits cleanly. The causal methods are exact, but their exactness recovers the wiring of the circuit without recovering what the variables mean.

Table 3 Mechanistic interpretability scored against the intervention oracle (Section 3.2). Ten methods on the six core games (oracle-as-method positive control = 1.0): faithfulness F with its 95% bootstrap CI, the plausibility proxy (Section 3), and the kind of recovery the oracle certifies. The causal core is exact by construction; the approximate and search-based methods lose recall; and the named-but-inert and present-but-unused methods sit in the danger zone (high plausibility, low faithfulness). Values read from the committed leaderboard [35].

Method	F (95% CI)	Plaus	Recovery
Activation patching [17]	1.00 $[1.00, 1.00]$	0.50	exact effect table
Causal scrubbing [43]	1.00 $[1.00, 1.00]$	0.50	exact data-flow preserved
Interchange interventions (DAS) [22]	1.00 $[1.00, 1.00]$	0.50	verifies a supplied alignment
Logit / tuned lens [26]	1.00 $[1.00, 1.00]$	0.50	exact intermediate value
NMF/PCA dictionaries	0.54 $[0.44, 0.65]$	0.75	named, partly inert
Attribution patching [27]	0.50 $[0.23, 0.78]$	0.90	approximate (linear surrogate)
ACDC [20]	0.47 $[0.18, 0.78]$	0.50	search (graph-correct, S=0.44)
Path patching [19]	0.25 $[0.03, 0.58]$	0.50	search (single-shot edges)
Sparse autoencoder [23]	0.21 $[0.02, 0.49]$	0.75	named yet inert (F=0.04 on Pong)
Linear probing (+ control) [25]	0.11 $[0.04, 0.17]$	0.85	present, not used
Oracle (positive control)	1.00 $[1.00, 1.00]$	1.00	exact intervention ground truth

Exact recovery of the wiring

The main result is simple. Activation patching recovers the true causal-effect table exactly. Across all six core games the recovered per-site effect equals the exact intervention oracle to the last bit (maximum absolute deviation 0.0), with site precision and recall both 1.0. On this machine, clamping a RAM cell to a donor value and re-running the real ROM is literally the oracle's own operation. So it cannot disagree with the oracle. Causal scrubbing inherits the same strength. The true data-flow hypothesis preserves behaviour under resample-ablation of everything outside it on all six games (preserved performance 1.0). A deliberately wrong hypothesis does not (0.52), and the two are discriminated on every game. Interchange interventions with distributed alignment search (DAS) verify the supplied alignment at interchange accuracy 1.0 against a mis-aligned control at 0.0. The logit lens reproduces the true intermediate value at fidelity 1.0 on every game. Four of the five causal methods sit at faithfulness 1.0, level with the oracle positive control (Table 3). Where a method's operation *is* an intervention, faithfulness is automatic on a fully observable machine. This is the first part of the problem, and it is solved here.

Approximate and search-based recovery

The causal account weakens as soon as a method replaces exact re-runs with an approximation or a search. Attribution patching, the gradient-linear surrogate for activation patching, correlates with the exact patch at only 0.50 and matches it inside the linear regime in just 0.19 of cases. Yet its recovered edges stay precise (edge precision 0.95, recall 0.70) because the linearisation is wrong about the size of an edge, not about whether the edge exists. Path patching, run in the IOI style as a single directed do-operation, recovers the directed edge set of the true routine at mean $F1 = 0.25$. On Q*bert it matches the empty direct circuit perfectly ($F1 = 1.0$). On Pong it recovers none of the six true edges ($F1 = 0.0$), because each receiver's value is re-derived within the horizon, so no one-step path carries a surviving signal. ACDC discovers the circuit automatically by resample-ablation edge pruning over a threshold sweep. It reaches mean $F1 = 0.47$ (ROC-AUC 0.70). A positive control re-runs the identical edge-restricted procedure under the exhaustive do-set and scores $F1 = 1.0$ on every game, so the deficit is a real measurement of single-shot recovery, not a broken scorer. These are the methods the field reaches for when the exact intervention is too expensive. On a machine where the exact intervention is cheap, these methods miss real structure. They are precise about what they find but silent about dependencies re-derived through other cells. The limit is recall, not precision. The gap to the positive control is the quantified cost of approximation against a circuit that is known.

Three separations from the oracle

This subsection presents the central result. ACDC on Breakout recovers the data-flow circuit at perfect $F1$ against the true routine, and the recovered graph is minimal at the algorithmic level (F= 1.0, M= 1.0). Yet that same graph, scrubbed and asked to regenerate the behaviour under held-out interventions, preserves only S= 0.44 of it (mean scrub-preserved performance 0.32 across games). A circuit can have the correct graph and still not reproduce the computed function. The missing object is not a denser or cleaner graph of the same type. Activation patching has already recovered the effect table exactly, with no tuning budget left to spend. A table of per-site effects is still not the bounce algorithm. One method is exact and the other recovers a perfect graph. Even so, neither reproduces the behaviour. A "methods underperform" explanation cannot account for this separation, because no method underperformed here.

The dictionary methods show the same gap from the opposite direction. A sparse autoencoder trained on the recorded RAM trajectory reconstructs the state almost perfectly (fraction of variance explained > 0.99 on Pong) and matches every verified game-variable cell it was asked to (matched fraction 1.0). By the metric the field uses to claim a feature "is" a concept, the SAE succeeds. By the oracle, it does not: ablating the matched Pong feature and re-running the real ROM gives causal faithfulness F= 0.041, with S= -0.336. When we remove the matched feature, the computation it is supposed to name does not change. Across the six games the SAE's mean leaderboard faithfulness is 0.21 against a method-tradition plausibility proxy of 0.75, squarely in the Section 3 danger zone. The NMF/PCA dictionary contrast repeats the pattern at lower resolution (matched-component fractions up to 0.75 beside near-zero causal use). This separation is the reverse of the first. The circuit was causal but behaviourally incomplete; the feature is perfectly named but causally inert. The two failures point in opposite directions.

Linear probing shows a third version of the same gap, the well-known probing trap. With control tasks (Hewitt & Liang's selectivity), probes decode game-concept cells well above chance (mean probe accuracy 0.89, mean selectivity 0.16), confirming that the information is present in the state. Yet the program does not necessarily use that information. Cross-referenced against the exact intervention oracle, the oracle flags 25 concept cells across the core games that the program never causally uses within the window. Of those, 6 are also decodable above chance, so a probe would have nominated them as encoded concepts. Breakout's score cell 84 sits in the larger set: it is present in the state and linearly associated with the displayed score, yet provably inert under intervention within the window. The oracle is strictly stronger than the probe. The probe asks whether a value can be read out. The oracle asks whether changing that value changes the output. Probing earns a high plausibility proxy (0.85) on a faithfulness of 0.11, the lowest faithfulness in the danger zone (Table 3). We state the caveat directly: these labels are imported externally (AtariARI, OCAtari) and only *verified* causally, never *discovered*. So DAS's alignment accuracy of 1.0 is guaranteed in advance, because it only confirms a variable we supplied. The probe tells us a concept is present. Only the intervention tells us the program uses it.

Exactness is necessary but not sufficient

Mechanistic interpretability is the clearest test of the paper's claim. Run against a known real circuit, the mechanistic toolkit splits into two groups. The first is a causal core that is faithful by construction: activation patching, causal scrubbing, DAS, and the logit lens, all at 1.0. Together with the intervention methods this causal and intervention family averages 0.66 faithfulness, the highest of the three traditions. The second is a recovery layer that is either precise but partial (attribution and path patching, ACDC) or present but inert (SAEs, dictionaries, probing). The oracle, not our judgement, decides the three separations. They appear exactly where the field's success metrics report success: graph-correct yet behaviour-incomplete (ACDC on Breakout F= 1.0/M= 1.0, S= 0.44), named yet causally inert (SAE on Pong matched 1.0, F= 0.041), present yet unused (cell 84). The faithfulness advantage that the causal and intervention family holds over the gradient methods is robust across all output regimes. The all-regime gap is 0.30 (95% CI $[0.23, 0.38]$, excluding zero). Restricted to the position-output regime alone the gap is 0.24, in the same direction, but at six games its interval crosses zero ($[-0.05, 0.32]$), so on position we report a directional trend, not a significant win. The separations above do not rest on either family mean, because each is a within-method contradiction on a single game. On a machine that gives faithfulness for free, faithfulness is necessary for understanding but not sufficient. The residual is the semantic gap: the wiring is recovered, but the meaning is not. Section 1.4 places this gap on the shared axes, and the Discussion treats it as the field's open problem.

1.4 Across the traditions: causal structure without meaning

We have now run three traditions on one machine. This section is organized not by the ranking among them, but by a boundary they share. Every method, faithful or not, returns the system's causal structure but not the meaning of its variables. Faithfulness is whether an explanation names the true causes. On the VCS we can measure it exactly. The intervention oracle of Section 3.2 re-runs the bit-exact program for every candidate cause. It returns ground truth that no real network can provide. That exact answer lets us do two things in sequence. First, we place all three traditions on the shared faithfulness-versus-plausibility plane. We read off the leaderboard and its danger zone, the known result that faithful and convincing are not the same. Second, we go further. We make the unfaithful methods faithful by turning on the differentiable port's bilinear sampler. Making every method faithful does not change the result, because faithfulness was never the dividing line. The harder part of the problem is the variable's meaning and the routine's algorithm. This is a separate object, and no method on this fully known machine recovers it. We call the distance to it the *semantic gap*. The leaderboard is an intermediate result; the semantic gap is the main result. The correctness triad we apply throughout — **F**aithful, **S**ufficient, **M**inimal — is defined formally in Section 3.4; here we only read its scores.

The faithfulness leaderboard

The leaderboard separates cleanly along the faithfulness axis. We read it as an exploratory cross-tradition ranking, not a single commensurable score, because the traditions measure different objects. We report the ranking with its robustness. We aggregate the 257 committed per-game records into the canonical 30 interpretability methods plus the intervention oracle as a positive control, 31 rows in all [35] (Fig. 2). The causal and intervention methods average a faithfulness of 0.664 against the oracle, while the gradient and correlational methods average 0.367, a gap of 0.297 across all output regimes. This all-regime gap is the robust headline result: its 95% bootstrap interval is $[0.23, 0.38]$ and excludes zero [35]. The position regime is the set of discrete sprite-position outputs whose naive gradient is provably zero under the hard forward map (Section 1.2). On this regime the gap points the same way, causal and intervention 0.486 against gradient and correlational 0.248, but at six games its interval crosses zero ($[-0.05, 0.32]$), so on position we report a directional trend and not a significant win. The ordering is the one the method matrix predicted (Section 3), now measured. Activation patching, causal scrubbing, interchange interventions, and the logit lens sit at the oracle ceiling of 1.0; the single-unit lesion follows at 0.99; the classical correlational analyses (tuning curves, Granger causality, and spike-word correlations) score near zero. The clearest single-method contrast is on the position regime. Activation patching reaches faithfulness 1.000, the oracle ceiling, on all six core games, while vanilla saliency, a widely used attribution baseline, reaches only 0.165, a per-method gap of 0.835 [44]. Hence, on a system where the answer is known exactly, an everyday attribution tool scores far below a causal method on the discrete outputs.

The all-regime separation survives reweighting, and where it does not we say so. Reading the robustness record [35], the causal-versus-gradient gap stays at 0.297 under both equal weighting by method and equal weighting by game, and its 95% bootstrap interval excludes zero in each case ($[0.23, 0.38]$). Yet three qualifications must be stated alongside this result. When the oracle-equivalent intervention methods are removed, leaving only coarser approximations, the gap shrinks to 0.018 and its interval includes zero, so much of the leaderboard's main contrast comes from methods close to the oracle operation itself. The gap is also regime-specific in a way that cuts against a simple "gradients fail" reading. On content outputs, where the naive gradient is valid, the gradient and correlational family is actually the higher of the two (0.637 against 0.523), so the content gap is -0.114 with an interval spanning zero. On the position regime the gap is 0.238 in the expected direction, but its interval also crosses zero ($[-0.05, 0.32]$) at six games. The conclusion that survives reweighting is the all-regime one: pooled over every output, causal and intervention methods are more faithful than gradient and correlational methods, and this holds under both weightings. The

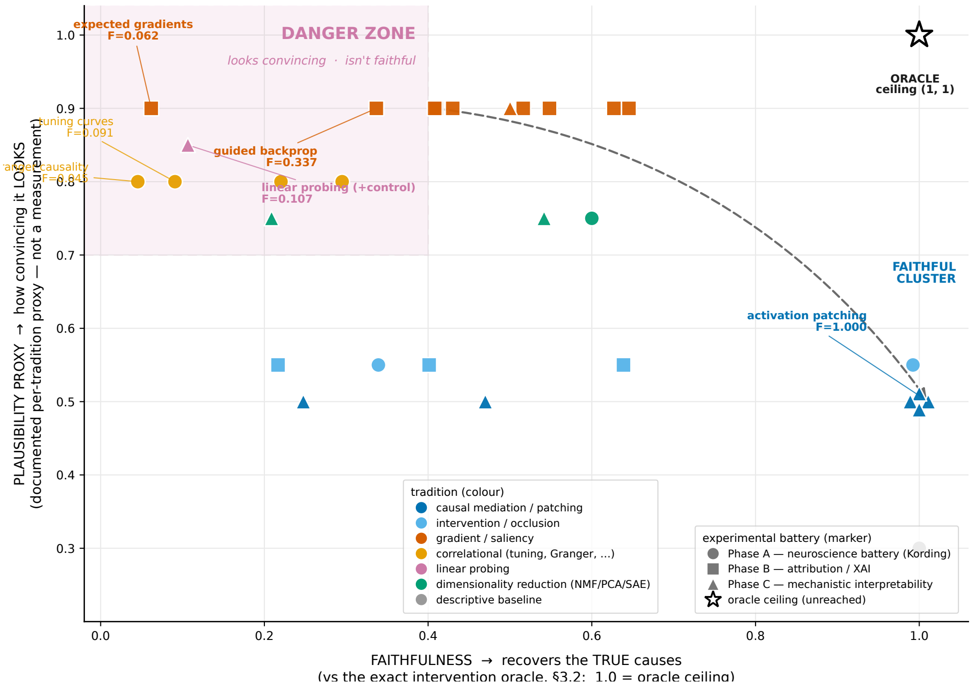

Fig. 2 Every method, every tradition, on the shared faithfulness-versus-plausibility plane. The plotted set is the canonical 30 interpretability methods + the oracle = 31 rows, each placed by its faithfulness against the intervention oracle (Section 3.2; X, exact ground truth) and by its documented method-tradition *plausibility proxy* (Y; Section 3). The plausibility axis is a tradition-level proxy, not a human-subjects measurement. The oracle positive control anchors (1.0, 1.0), a ceiling no interpretability method reaches. Pooled over all outputs the causal and intervention family is more faithful than the gradient and correlational family (0.66 against 0.37, a gap of 0.30 whose 95% interval excludes zero). Note that several gradient and correlational methods sit low on faithfulness yet *higher* on the proxy — the *danger zone* of high apparent plausibility at low faithfulness. Among the gradient methods the danger-zone offenders are expected gradients ($F = 0.06$) and guided backprop ($F = 0.34$) at plausibility 0.9; vanilla saliency is not one of them, since its all-regime faithfulness is 0.42. Every plotted value reads from the committed leaderboard, none re-run or estimated; per-method values with 95% confidence intervals are in Tables 1–3 and the supplement.

position-regime advantage is directional and consistent with the mechanism, but not statistically established at this sample size, and we do not claim it as significant. This is the cross-tradition analogue of what neural-network benchmarks report when faithfulness is scored against a constructed ground truth [45, 46], with the difference that here the ground truth is an exact intervention on a real software artifact rather than a synthetic or pseudo-label.

The danger zone

If the faithfulness axis matched how convincing a method looks, it would tell us nothing new. It does not match. Plotted against the plausibility proxy of Section 3, the points split into a safe diagonal and a dangerous corner. The methods the field uses first sit near zero faithfulness while having the highest plausibility-proxy values. Expected Gradients tops the danger list at plausibility 0.9 against faithfulness 0.062; linear probing follows at 0.85 against 0.107; A3 tuning at 0.8 against 0.091. This inversion is the key finding: a map that is weak or empty on the position regime looks *more* convincing than the exactly correct one. We state the caveat plainly. The plausibility axis is a documented method-tradition proxy assigned per tradition before the faithfulness scores were read, not a human-subjects measurement, and we report it as such to avoid overclaiming the danger zone. Still, the faithfulness axis is exact, and the ranking it inverts is what the field tends to trust, so the hazard is real even at the proxy's coarse resolution. The danger zone is a quantitative result, not a speculation.

Making the unfaithful faithful

A reader could accept the danger zone and still dismiss it by blaming the gradient. The saliency methods score zero on the position regime because the hard forward map reads a sprite's position through an integer index, so the gradient is zero almost everywhere (Section 1.2, Prop. 1). On this reading, the danger zone is a technical artifact a smoother forward pass would repair. Once repaired, the gradient methods would become faithful and the concern would disappear. The differentiable port lets us run exactly that repair and test the claim. We turn its bilinear sampler on, replacing the integer index read by a triangular-kernel occupancy whose sprite column is a continuous function of the position byte. This restores a non-zero gradient through the position computation without altering the forward output. The same surrogate that recovers the joystick gradient in Section 1.2 now flows through the position read.

The repair restores a non-zero gradient, but its faithfulness stays low and mixed (Fig. 3). On Pong, vanilla saliency, gradient $\times$ input, SmoothGrad, and expected gradients all rise from the provable zero to faithfulness 0.791 against the oracle; on Q*bert vanilla saliency, gradient $\times$ input, SmoothGrad, and integrated gradients rise to 0.529 (committed sampler record). The rise is the exception, not the rule. Averaged over the six position games the restored faithfulness is only about 0.22 for the SmoothGrad-family methods and about 0.09 for integrated and expected gradients, because the gradient becomes non-zero on just one or two games. Integrated gradients with a zeros baseline still vanishes on Pong, the static-sprite games offer no position to restore, and on Space Invaders the restored gradient lands on a non-active cell, so it does not move. So turning the sampler on does not make the gradient methods faithful. It only removes the objection that their position score is a pure artifact of a non-differentiable read. A faithful-capable substrate is not the same as a faithful method.

But this does not give understanding. The restored gradient on Pong concentrates entirely on the single RAM byte that holds the paddle's position, `ram[54]`, which the oracle independently confirms is a true cause. The method now finds the right cause and attributes nothing to the wrong ones. Yet it still recovers no meaning, exactly as when the map was empty, and exactly as no other method in the study recovers any (same committed record). A faithful attribution to `ram[54]` names a byte the program reads. It does not say that the byte *is* the paddle, that reading it *is* tracking, or that the routine computes a bounce. The only semantic label in play is the imported annotation

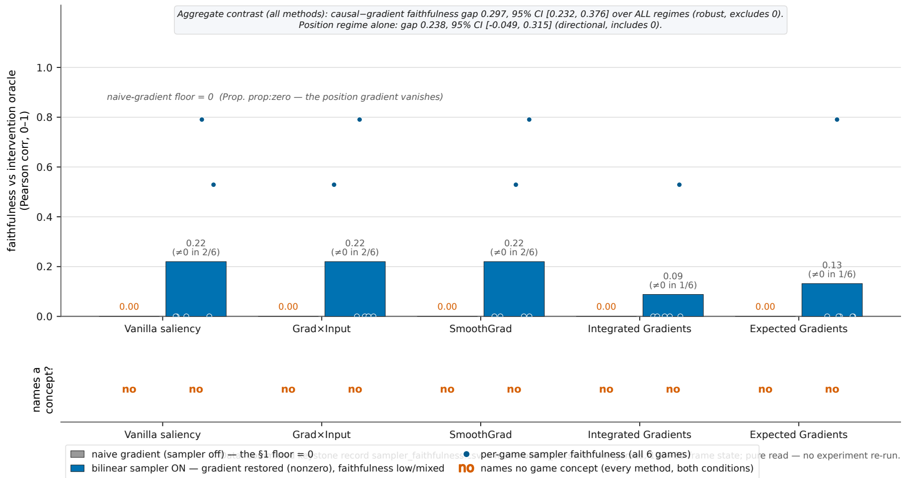

Fig. 3 A differentiable substrate restores a non-zero gradient, but the restored faithfulness stays low and mixed, and no method names a game concept. Top: on the position regime the naive gradient is provably zero (Prop. 1), so the saliency methods reach faithfulness 0 (naive bars); turning the differentiable port's bilinear sampler on restores the gradient through the position read and lifts faithfulness to 0.791 on Pong and 0.529 on Q*bert for the methods with a moving tracked sprite (sampler bars), the restored gradient concentrating on the true position byte (`ram[54]` on Pong, oracle-confirmed). Note that the rise is confined to one or two games: averaged over the six position games the restored faithfulness is about 0.22 for the SmoothGrad-family methods and about 0.09 for integrated and expected gradients, so a faithful-capable substrate does not make the gradient method faithful. **Bottom:** semantic recovery is a yes/no question, and the answer is no in both conditions and for every method: an attribution map names a cause but emits no game concept; the only semantic labels are imported (Section 3.2) and used to check localization, never produced. The rise is reported only where a tracked sprite exists: integrated gradients with a zeros baseline vanishes on Pong, static-sprite games have no position to restore, and Space Invaders' restored gradient lands on a non-active cell. Values read from the committed sampler record; none re-run.

we use to *check* where the attribution landed, never one the method *produced*. Even where the restored gradient is faithful, it carries no meaning. This result sets up the rest of the section: the gap we measure next is not the unfaithful methods' deficit, because restoring the gradient does not close it (Fig. 3).

Where the leaderboard stops — the semantic gap

Here the paper turns. Take the best case for the audit: a method that scores 1.0 against the oracle. We ask whether it gives an understanding of what the program does. It has not, and we are careful about what that claim is and is not. We do *not* claim that no method can recover meaning; that is the unbounded statement, the subject of the companion design-recovery work, which we have not run here. We claim the bounded thing the oracle can prove: even when a method recovers the true causal structure exactly, what it returns is the causal structure — a causal-effect map, a data-flow graph — and not the *semantics*, the variable's meaning or the routine's algorithm. The missing object is a different kind of thing. It is not just a poorer version of the causal structure. Following the triad of Section 3.4, we operationalize the gap through three separations, each adjudicated by the oracle: faithful-not-sufficient, named-not-causal, and decodable-not-used. They diagnose distinct semantic failure modes; they are not a complete theory of semantics. The three separations show the failures pointing in opposite directions (Fig. 4).

The first separation is graph-correct yet behavior-incomplete: high F and M, low S. Automatic circuit discovery (ACDC) [20] on Breakout recovers a data-flow graph that is, by the exact data-flow oracle, perfectly correct and perfectly parsimonious, $F = 1.0$ and $M = 1.0$ [35]. Yet the discovered circuit alone, with its non-circuit parents scrubbed, reproduces the clean candidate-cell readouts only 44% of the time, $S = 0.44$. A correct graph of which cells feed which other cells is not the function those cells compute. The faithful side shows the same. Activation patching [17] recovers the exact per-site causal-effect table on every one of the six games [35], and a table of effects, however exact, is not the bounce algorithm that produced them. No tuning budget is left to spend here, because the effect table is *already* exact, so the residual to the algorithm cannot be a worse fit closed by a better hyperparameter. It is a missing object of a different type, and that is why the gap is not merely "methods underperform."

The second separation is named yet causally inert, the reverse failure: high matched fraction, low causal F. A sparse autoencoder [23] trained on the recorded Pong state matches every one of its candidate game-concept variables, a feature-to-variable matched fraction of 1.0, a perfect semantic alignment by the field's usual yardstick. By the oracle, those same matched features are nearly causally inert on Pong: faithfulness $F = 0.041$ and a *negative* sufficiency $S = -0.336$ [35]. A feature that looks like the ball's x is not the same as the mechanism the program uses to compute with the ball's x. The circuit was correct but incomplete; the dictionary is named but causally wrong. The gap runs in the opposite direction on the same axis. An account of methods simply scoring too low cannot explain this. Recent SAE evaluations report the same proxy-versus-use gap on neural language models [29]; the oracle here turns that observation into an exact causal-use test on a known system. We guard against reading semantic success into an aligned variable. The interchange-intervention alignment (DAS) [22] reports an aligned-variable accuracy of 1.0, but that 1.0 is true *by construction*, verifying a *supplied* variable rather than discovering one. A method that confirms a given concept is not a method that found the meaning.

The third separation is decodable yet unused, the present-versus-used trap, where the oracle is strictly stronger than the probe: high probe accuracy and selectivity, low intervention effect. Linear probing with control tasks [25] decodes game concepts from the Breakout state at a mean accuracy of 0.72 with positive selectivity over its control; the information is demonstrably *present*. Yet the exact oracle flags a score-related cell that is *present but unused* (cell 84): re-derived within the horizon, its value has no downstream effect on the output [35]. A concept can be linearly readable yet play no part in the computation, exactly Hewitt and Liang's control-task warning; decodability is necessary, never sufficient. Probing answers whether the concept is present. The oracle answers whether it is used. Only the second tells us how the program works. Linear probing's leaderboard faithfulness of 0.107 paired with a plausibility proxy of 0.85 is its place in the danger zone, and this cell is the mechanism behind that placement.

Read together, the leaderboard plane of Fig. 2 and the failure taxonomy of Fig. 4 make the same point from three sides, and the triad's axes are not redundant: a method can be faithful and minimal yet not sufficient (ACDC on Breakout, $F = M = 1.0$, $S = 0.44$); it can match a named concept perfectly yet be causally inert (the SAE on Pong, matched fraction 1.0, $F = 0.041$); it can flag a concept the program never uses (cell 84, present but unused). The claim does not rest on any single low number; a lone low S could be a tuning artifact. It rests on two things. First, the gap appears where tuning cannot help: the effect table is already exact. Second, the failures run in opposite directions on the same axis. A "methods are not good enough yet" account cannot explain this two-sided result. We restate the honesty contract here, because this is the easiest place to overread the results. We have *not* recovered the program's meaning, algorithm, variables, or design from scratch. Every semantic label we used is external, imported from AtariARI [32] and OCAtari [33]. The substrate only *causally verifies* a supplied label, it does not *discover* the concept, and the provenance record marks each label as imported rather than recovered. So the faithful methods recover the causal structure. The meaning is a separate thing, and we have measured the distance to it. Recovering that meaning is a different problem, and it is the companion paper's, not this one's.

A note on aggregation closes the section, because two committed records report the sparse autoencoder differently and a careful reader will see both. The leaderboard score [35] aggregates over the per-method records, which for the SAE gives faithfulness 0.209 with a 95% bootstrap interval of $[0.02, 0.49]$ that brackets it; the figures plot this value (Supplement S5 records both aggregations). The per-game Pong record used in the named-not-causal separation above is a different, finer object, $F = 0.041$ on that one game. We keep the two distinct: the 0.209 is the leaderboard aggregate, the 0.041 is the single-game illustration. Throughout the paper, the leaderboard and its figures use the per-method aggregation as canonical, so text, table, and figure agree.

2 Discussion

We applied the interpretability toolkit to a machine whose ground truth is computable. This let us measure two things at once. On this system the true cause of any output is recoverable by exact intervention. The causal and intervention methods are faithful, while the field's familiar gradient and correlational tools are not. The leaderboard separates them by faithfulness, the one axis we can score exactly (Fig. 2). Pooled over all outputs the causal and intervention methods reach 0.66 against the gradient and correlational methods' 0.37, a gap of 0.30 whose 95% interval excludes zero. On the discrete position outputs a causal method reaches 1.000 where standard saliency reaches only 0.165, on a machine whose answer is known exactly. Yet faithfulness is the easier of the two measurements. Even a method at the oracle ceiling returns only the causal wiring: an effect map and a data-flow graph. It does not return the semantics: the meaning of a variable or the algorithm of a routine. We call this residual the *semantic gap*, and it is a number we can measure.

We score three traditions with one common oracle: a neuroscience battery, the attribution toolkit, and mechanistic interpretability. These traditions are rarely studied together, but the oracle places them on one scale. Jonas and Kording warned that an analyst armed with systems neuroscience could fail to understand even a microprocessor, because describing structure is not the same as explaining how it works [4]. We make that a measurement. The classical battery recovers rich structure yet scores low against the known mechanism. The naive gradient is provably zero on the discrete position regime, and only the causal operators reach the ceiling. Mechanistic interpretability has until now been calibrated against synthetic or compiled ground truth, transformers with a planted circuit by construction [54, 55], ground-truth attribution benchmarks that plant the important features [46, 56], and circuit- and variable-localization benchmarks that curate their answers in neural language models [45]. We add the missing case: a real, human-written, deployed artifact that was never designed to be interpretable. Its ground truth is independent of the artifact, not built into it. Our subject is not the Atari environment itself, which the Arcade Learning Environment and its faster and object-centric descendants already supply [57–60]; it is the bit-exact, differentiable intervention oracle layered on it. So the leaderboard calibrates these methods against ground truth on a system the field did not design to be interpretable.

The same oracle also reveals a limit: a high leaderboard score cannot show meaning. This limit is the subject of this paper. A perfect faithfulness score certifies that a method named the true causes. It does not certify that the method recovered what the program means. We observed this limit in three cases (Fig. 4). On Breakout a circuit is graph-correct and minimal ($F = M = 1.0$) yet reproduces behavior only 44% of the time ($S = 0.44$). On Pong a feature dictionary matches every named variable (matched fraction 1.0) while being nearly causally inert ($F = 0.041$, $S = -0.336$). And a concept is linearly decodable yet, by the oracle, never used in the computation. The gap appears in two opposite forms: correct-but-incomplete, and named-but-wrong. A simple "the methods are not good enough yet" story cannot produce both.

We tested the strongest version of that story and it failed. The faithfulness gap could be read as an artifact of one design choice. The discrete position read-out makes the naive gradient vanish by Prop. 1. So the gradient family scores a provable zero on the position regime, and the failure could be dismissed as a fixable technical artifact. We removed that artifact. Turning the port's bilinear sampler on restores a usable gradient through the position index. On the games with a moving tracked sprite the restored gradient becomes faithful. On Pong it reaches 0.791 for vanilla saliency, grad×input, smoothgrad, and expected gradients. On Q*bert it reaches 0.529 for vanilla saliency, grad×input, smoothgrad, and integrated gradients (Fig. 3). The restored gradient concentrates on the position byte that drives the sprite, `ram[54]` on Pong; it finds the right cause. Yet the rise is confined to one or two games. Averaged over the six position games the restored faithfulness is only about 0.22 for the SmoothGrad-family methods and about 0.09 for integrated and expected gradients. So a faithful-capable substrate does not make the gradient method faithful in general. Where it does become faithful, none of these methods recovers any meaning, exactly as before, and exactly like every other method in the study. Even where the gradient is corrected, it identifies the cause but no meaning. So the meaning was never present in the wiring for a method to recover.

The improvement occurs only in specific cases, and that pattern is itself evidence. It is not a universal rise. It rises only on the games with a moving tracked sprite the gradient can flow to. Integrated gradients with a zeros baseline still vanishes on Pong. The static-sprite games show no rise, and so does the case where the restored gradient falls on a cell the program does not currently read. Where a position byte exists and moves, the sampler makes that gradient faithful. We gave the faithfulness explanation every case it could win. None of it produced a single semantic label (Fig. 3).

What understanding would require

Faithfulness stops short for a reason that Marr's levels of analysis make clear [61]. A faithful attribution lives at the implementational level, where cells and registers carry the signal. Understanding lives one level up, at the algorithmic level, in what function the program computes. A correct data-flow graph is an implementational fact. The bounce algorithm it implements is an algorithmic fact, and the two are objects of a different type. Lazebnik made the same point: one can describe a system in complete mechanistic detail and still not understand it [62]. We turn that observation into a measured quantity. The distance from a faithful effect table to the algorithm is the sufficiency gap $1 - S$, and on an already-exact effect table it cannot be closed by a better hyperparameter. Our operational definition of interpretability, ground-truth recovery scored by the oracle, is one of several in circulation; others set it as a design target for new models rather than an audit of a system already built [63].

Software engineering gives the clearest version of this requirement. IEEE 610.12 defines software as programs plus their documentation and data, the specification, the design, the intent [34]. A faithful circuit recovers part of the program. To understand the system you must also recover its documentation: what each variable is for and what each routine should do. The sampler-on result tells us why no internal method can reach that object. The meaning of a variable is not a causal fact about the variable. It is a fact about the behavior the design serves, and that behavior is fixed outside the program code, in the manuals and the data the program was written against. A method that reads only the mechanism, however faithful, reads only causes. A cause becomes a meaning only against an external reference: the system's documented behavior. No perturbation of the chip can reach that reference. In this study the only semantics present entered from outside: every T3 label was imported from AtariARI and OCAtari and then merely verified to be causally live, never discovered by a method. Hence the gap is not an artifact of weak methods but a property of the question. Faithfulness is necessary for understanding but not sufficient, because understanding needs an external behavioral reference that an internal method does not carry. We state clearly that this paper defines and measures the gap. Closing the gap means recovering the program's documentation and design from the bare machine. That is the reverse-engineering goal, and the task of the companion design-recovery paper (Paper 4). That paper measures the distance to that goal rather than clearing it. Whether a recovered concept matches how the system is used is the companion behavior paper (Paper 3). The learned agent, where there is no disassembly to check against, is Paper 5.

Probing reads the same result from the data side. It shows that a concept is present in the state, but not that the concept is used. This is the control-task warning of Hewitt and Liang [25]. Our intervention oracle is strictly stronger, because it asks whether the byte causes the output, and it found a present-but-unused score cell the program never reads. Decodability is presence; use is an algorithmic fact. Understanding needs use, and only the oracle supplies it. At base, the three traditions are one toolkit under different names. Tuning curves match feature visualization, lesions match ablations, and Granger causality is the correlational predecessor of causal attribution. Each pair shares a failure mode, so a result measured on one transfers as a caution to the others. A benchmark built on an Atari chip thus speaks to systems neuroscience, explainable AI, mechanistic interpretability, and software verification at once.

Why a 1977 chip is a fair screen

We address the obvious objection. The VCS is deterministic, hand-coded, and small, while the systems these methods target are learned, distributed, stochastic, and large. We do not ask a score here to predict a transformer. The benchmark is a necessary-condition stress test, not a sufficiency predictor. Here a method has perfect ground truth, full observability, exact interventions, and no learning to complicate the result. If it still fails, that failure is strong evidence against any claim that the same method reliably recovers causal mechanisms in harder, less observable systems. Passing here is necessary but not sufficient; failing here is informative. The same logic suggests, rather than proves, a reading of the semantic gap. If even the strongest causal account a fully observable machine permits stops short of meaning, the result suggests that faithfulness alone should not be treated as understanding in less observable neural systems, where the circuit itself is uncertain and no clean ground truth is on hand. We measure $1 - S = 0.56$ on an exactly recovered circuit on Breakout ($1 - 0.4444$; the ACDC family mean is $1 - 0.31 = 0.69$), and we do not extend that number to neural networks as a proven floor; the precise scope, and what this paper does not claim, are set out in the Supplementary box on what we do and do not measure (S6).

The chip also exhibits the properties that make real systems hard to interpret. There are aliased, polysemantic RAM cells reused across routines, a hand-coded analogue of superposition. There is computation by racing the electron beam, which is distributed and temporal. And there are registers that mean different things in different contexts. Figure 5 maps each VCS property to the neural-network difficulty it does and does not capture, so that a property we can prove on the VCS is one to suspect in neural networks. We state plainly what the chip does not capture: learned features, scale, and stochasticity are absent. The learned agent that adds all three back is Paper 5.

What interpretability needs

Three recommendations follow. The field should prefer causal over correlational evidence, because the methods it reaches for first carry the highest plausibility proxy and the lowest measured faithfulness, and a saliency map provably empty on the position regime can look more convincing than the exactly correct map. It should validate on known systems before trusting on unknown ones, since a method that cannot pass where the answer is computable cannot be audited where it is not. Most important, it should stop treating faithfulness as understanding. The open problem for explainable AI is not a better saliency map; it is the semantic gap. We confirmed this directly: we closed the faithfulness gap for the gradient methods, and the meaning still did not appear. The field does not need an instrument that reads the mechanism more sharply. It needs one that supplies the reference the mechanism does not contain: the system's behavior, documented for software in its manuals and data. Faithfulness is the part we can now measure, and on causal methods we can achieve it. Meaning remains unsolved, because no internal method carries the external reference it needs.

We have measured a gap, not closed it, and several limits bound what we may conclude. We did not run semantic recovery, because Phase E, documentation and design recovery, is specified but not executed. Our claim is therefore the bounded one: on exact ground truth, faithfulness certifies the causal structure, and that structure alone is not the semantics. We do not make the unbounded claim that "no method can recover meaning," which is Paper 4's charge. The sufficiency and minimality axes are defined against the oracle, and the T3 semantic labels are external, imported from AtariARI [32] and OCAtari [33] and only causally verified, never discovered. A single low sufficiency could be a tuning artifact. So the claim does not rest on any single number. It rests on three facts: the gap appears where tuning cannot reach it; it runs in both directions; and it survives the sampler-on test that makes the gradient methods faithful on games with a moving tracked sprite. We do not read that test as universal. The rise occurs on Pong and Q*bert, but not on the static-sprite games, not on a zeros-baseline integrated gradient, and not where the restored gradient falls on a cell the program does not read. We report it as such, not as a law over all methods and games. The plausibility axis is a documented method-tradition proxy, not a human-subjects measurement, and we report it at that proxy's coarse resolution. Our scores hold within Paper 1's validated conformance window. Outside that horizon the substrate is not certified bit-exact, and we keep every scored claim inside it. The subject is one old machine with no learning, which is the necessary-condition framing once more: a stress test, not a forecast.

We end with the broader implication. Because the substrate is differentiable, it tells us which gradient explanations are trustworthy and where they vanish. The content path carries an honest gradient while the discrete index does not. So a saliency map over sprite position is empty for a structural reason, not by chance. This statement about gradients generalizes to any system where a discrete decision follows a continuous one. We believe a fully known, differentiable testbed is valuable for exactly this reason: it lets the community calibrate its instruments before deploying them where calibration is impossible. The hard case remains the learned agent, where there is no disassembly, no specification, and no engineer's intent to recover, and where the semantic gap is likely to be larger. Closing this gap may take the next decade of interpretability research. The goal is to turn a faithful account of a learned system into an understanding of what it has learned. We have not solved this problem. We have defined and measured it, which is the necessary first step.

3 Methods

Our method rests on a single idea. We study an interpretability question on a machine whose answer we can compute exactly. We describe in turn the substrate that makes the ground truth obtainable, the oracle that delivers it, the tiered semantics we attach to it, and the correctness triad that scores understanding as numbers. The per-method setups live in the Supplementary Information. Fig. 1 sketches the pipeline.

3.1 The differentiable, bit-exact VCS substrate

The subject of this study is the Atari Video Computer System (VCS) itself, not a learned agent. It is a real, complex artifact never designed to be interpretable. Yet we can measure its every causal dependency exactly. Paper 1 built it, a bit-exact, differentiable port of the 1977 VCS. The port is validated bit-for-bit against the `xitari` C++ reference on all 64 games of the Paper-1 roster, in random-access memory and rendered screen, within the validated conformance window [30, 31]. We use it in two interchangeable, cross-checked realizations, `jutari` (Julia) and `jaxtari` (JAX/Python).

The substrate grants four capabilities that no neural network offers for free. The first is an exact forward map,

$$y \;=\; \mathrm{VCS}\big(\mathrm{ROM},\, \mathrm{inputs}[0{:}t],\, s_0\big), \tag{1}$$

that is, the output y at step t, a pixel, a score, or a game event, is a deterministic function of the cartridge bytes, the inputs up to t, and the initial state s_0. Re-running reproduces y to the bit. The second is exact interventions. We set any cause u, a ROM byte, a RAM cell, a CPU/TIA/RIOT register, an opcode, or a joystick input, to any value and re-run Eq. (1) deterministically. This realizes the causal operator $\mathrm{do}(u)$ on a real machine rather than an assumed world model. The third is full state observability. We record every bit of RAM and every register along the trajectory, so the "activations" we interpret are the program's own variables. The fourth is a content-path gradient $\partial y/\partial u$ through a straight-through estimator (STE) formulation whose forward pass is bit-exact to the hard machine at any finite temperature [64, 65], batched on the GPU. A neural interpretability study works with three things: the inputs, the state, and the outputs. Our substrate gives us all three.

We respect one limit throughout. Paper-1 bit-exactness holds within a bounded frame window on a fixed action stream, on the order of 30 frames of RAM and 60 frames of screen. Every scored claim is taken inside that validated horizon. Long-horizon end-to-end gradients are not needed for our single-step metrics, so we do not assume them.

3.2 The ground-truth oracle

The oracle is central to the method. For any output y and cause u, it returns the true effect of u on y. We score every method against it. Its primary construction is the exact intervention oracle: we intervene on u by occlusion, clamping, replacement, or resampling, re-run the machine deterministically, and record the change in the output,

$$\Delta y(u) \;=\; \mathrm{VCS}\big(\ldots,\, \mathrm{do}(u{:=}v'),\, \ldots\big) \;-\; y. \tag{2}$$

The true causal effect of a cause is the difference its edit makes to the output. We measure it by running the altered machine. We assume no world model and no smoothness. The collection $\{\Delta y(u)\}$ over all causes is the true causal map of y, the reference for the faithfulness axis of every method.

The output y comes in five regimes. The regime fixes what a method must predict and whether a gradient is even meaningful. A *content* output is a register, color, or graphics-bit value that flows smoothly into the pixel, where the naive gradient is valid. A *position/index* output is a sprite, missile, or ball coordinate set by strobe timing. Here the naive gradient vanishes by Prop. 1, and only the bilinear-sampler surrogate or the intervention oracle applies. An *event* output is a discrete state change, a collision latch or a life lost, also gradient-zero. A *score* output is the game's running counter, valid for the gradient only where it varies continuously. A *pixel* output is the rendered framebuffer value. It is content-valid where it tracks a smooth cause and index-zero where it tracks a position. The Supplement lists, per game, which outputs each method is scored on and the per-output gradient verdict.

The oracle uses several kinds of intervention. We list each one, because the kind limits what a score means. Occlusion and clamping are surgical but *off-manifold*: they can set a byte to a value the running program would never write. Replacement from a donor trajectory is *on-manifold*: the substituted value is one the machine actually produced elsewhere. Resampling draws the cause from its own marginal over the trajectory. Counterfactual edits are valid under the emulator by construction, since any byte assignment yields a well-defined deterministic re-run. Off-manifold interventions are still valid for oracle scoring. The oracle measures a causal effect, not a likelihood. We ask what the output would be if u were v'. The bit-exact machine answers this exactly for any v', on- or off-manifold. Where a method's own claim is about typical behavior rather than causal dependence we fall back to the on-manifold donor and resampling variants and flag the choice in the Supplement.

The content-path gradient $\partial y/\partial u$ is the second way to build the oracle. We read it through the differentiable substrate. It is fast and dense, and it agrees with the intervention oracle on the content path, the register, color, and graphics-bit values that flow smoothly into the pixel. Yet, one important limit applies to the whole study. The straight-through gradient routes to the content path only. So a discrete index, a sprite's screen position the VCS sets by the timing of a strobe write, has a naive gradient that vanishes under the hard forward map. We make this precise.

Proposition 1 (Zero naive gradient for discrete-index outputs)**.** *Let an output y depend on a cause u only through a discrete index $n(u) = \lfloor g(u) \rfloor \in \mathbb{Z}$, the integer count of color clocks between a strobe write and the position-counter reset that the VCS uses to place a sprite, missile, or ball, with g continuous, piecewise linear, and non-constant (strictly monotone on each linear piece). Under the hard (bit-exact) forward map the rendered output is $y = h(n(u))$ for a fixed render function h. Then $n(\cdot)$ is locally constant wherever $g(u) \notin \mathbb{Z}$, so $\partial y/\partial u = 0$ at every such point, and y is discontinuous on the set $\{u : g(u) \in \mathbb{Z}\}$, which is measure zero precisely because g is* *strictly monotone on each piece, and there the gradient is undefined. Hence the naive gradient is zero wherever the derivative exists: a tiny change in u cannot move the sprite by part of a pixel. Any method that reduces to this naive gradient inherits the failure on position, index, and event outputs.*

The proposition states only the hard-map limit; it does not say that no useful gradient exists. Two surrogate constructions of Paper 1 [30] are explicit exceptions and are *not* covered by it: the straight-through estimator, whose forward pass stays bit-exact to the hard machine at any finite temperature while routing a non-zero gradient along the content path [64, 65], and the bilinear sampler that interpolates the index boundary to recover a usable position gradient [66]. The companion paper constructs both and validates the bit-exactness this proposition assumes. We import that construction and prove the vanishing inline, so the present paper stands alone. We therefore avoid any blanket claim that "gradients are provably zero." We use instead the scoped claim of Prop. 1: the *naive* gradient for discrete position/index outputs is zero under the hard forward map, and methods that reduce to it inherit that failure. Surrogate and bilinear-sampler gradients are exceptions.

We make the faithfulness audit as strict as possible. The position regime leaves one objection open, and we close it. A reader might object that the naive zero of Prop. 1 is just a fixable artifact. So we fix it: we turn on the Paper-1 bilinear sampler. The sampler interpolates the index boundary of Prop. 1 in the spatial-transformer manner. This makes the discrete position output a continuous function of the position RAM byte rather than a step. So the gradient methods receive a non-zero surrogate position gradient where the hard map gave them nothing [66]. We score both conditions, the naive forward and the sampler-on forward, against the same intervention oracle of Eq. (2). For each candidate cause the per-cause attribution is correlated, by the raw Pearson ρ of Section 3.4, against the oracle's true $|\Delta\,\text{position-pixel}|$. So the two conditions differ only in the surrogate the method is handed, not in how it is judged. Alongside faithfulness we record a second, deliberately stricter property. The *semantic recovery* of a method is a yes/no question: does the method, on its own, emit a game-concept name, the word "ball" or "missile" attached to the byte it points at, rather than a cause that an external label only names later? An attribution map returns weights over causes and never a name, so for it the answer is no by construction: it recovers no meaning. Faithfulness and semantic recovery are independent, and they are of different kinds. Faithfulness is a graded score for whether the method names the right cause. Semantic recovery is a yes/no property: whether it names any meaning at all. The keystone experiment of Section 1.4 reports both for the naive and sampler-on conditions on the position regime.

Hence, for content outputs the gradient is a legitimate companion oracle, while for position, index, and event outputs the intervention oracle of Eq. (2) is the sole ground truth and every gradient-based method is scored as a method under test. This position regime is where the sharpest separations in our results appear. Where the two overlap we report their correlation and flag, rather than hide, the points at which they disagree.

3.3 Tiered ground truth and how we obtain it

The ground truth comes in three tiers. The tiers are not just bookkeeping; they support the paper's central argument. Tier 1 (T1) is the exact causal map of Eq. (2), derived internally from the substrate and available for all 64 games. Tier 2 (T2) is the hardware roles and the read/write data-flow graph, also exact and internal across all 64 games. T1 and T2 describe what the machine does internally, and need no external label.

Tier 3 (T3) is different in kind. This difference matters for our argument. It attaches a game-concept name to a cell or feature, "the ball's x position" or "the score." These names cannot be computed from the substrate alone. They are facts about what the program *means*, which is separate from what it *computes*. We import T3 from two public sources and make it causal. AtariARI provides labels for 22 games, read from commented disassemblies, with raw RAM coordinates not aligned to the rendered position [32]; OCAtari covers more than 40 games and, in its RAM mode, applies the render-time offsets that fix that misalignment [33]. We import both as candidates and verify each causally by setting the byte, re-rendering, and confirming on the bit-exact framebuffer that the named object moves to the predicted place. This upgrades a correlational label to a verified-causal one and auto-corrects the offsets; further labels we discover by RAM-to-framebuffer correlation and intervention. The headline core set is six games, Pong, Breakout, Space Invaders, Seaquest, Ms. Pac-Man, and Q*bert, each carrying both sources, all in the Paper-1 bit-exact roster. Together they span paddle-and-ball, shooter-with-shields, resource-meter, multi-agent-pursuit, and discrete-grid logic. The remaining games form a breadth set scored on T1 and T2 only.

We are careful about what this gives us. All T3 labels are external and only verified, never discovered from scratch. So a method that matches a supplied concept has aligned to a given meaning, not recovered it. This is exactly why a probe can show that information is *present* without showing it is *used* [25]. Our intervention test asks whether the byte *causes* the named object to move, so it is strictly stronger than decoding it. Recovering the program's documentation and design from the bare machine is a separate problem, the subject of the companion paper.

3.4 The $\mathbf{F} \wedge \mathbf{S} \wedge \mathbf{M}$ correctness triad

A faithful attribution is not yet an understanding. To measure understanding, we need an operational criterion. We adopt one in the spirit of Marr's levels of analysis [61] and of Jonas and Kording's warning that finding structure is not the same as explaining it [4]: each axis is first a continuous score in $[0,1]$, and the boolean "right" is a thresholded reading of those scores. We score an explanation $\hat{E}$ of how an output y arises by three numbers, $\mathrm{F_{score}}(\hat{E})$, $\mathrm{S_{score}}(\hat{E})$, $\mathrm{M_{score}}(\hat{E}) \in [0,1]$, and call it *right* when all three clear their thresholds,

$$\mathrm{right}(\hat{E}) \;=\; \mathbf{1}[\,\mathrm{F_{score}} \geq \tau_{\mathrm{F}} \,\wedge\, \mathrm{S_{score}} \geq \tau_{\mathrm{S}} \,\wedge\, \mathrm{M_{score}} \geq \tau_{\mathrm{M}}\,], \tag{3}$$

with $\tau_{\mathrm{F}} = \tau_{\mathrm{S}} = \tau_{\mathrm{M}} = 0.5$ throughout. Results report the continuous scores; the predicate is a derived summary, not the primary quantity. All three are defined against the oracle, not against any method under test, which keeps the criterion non-circular.

Faithfulness asks whether the claimed causes are the *true* causes, scored against the intervention oracle of Eq. (2). The per-record metric depends on the explanation's output type, so $\mathrm{F}_{\text{score}}$ is a single number assembled from three type-matched primitives. The committed score is the bare effect-agreement primitive, not a rescaled one. For a real-valued attribution map we report the Pearson (or Spearman, for ranked outputs) correlation $\rho(\hat{E}, \{\Delta y(u)\})$ between the attribution and the true causal effects, scored *raw*. A method whose attribution carries no causal signal therefore lands at $\rho = 0$, not at the 0.5 of an affine $\frac{1}{2}(1+\rho)$ remap. This is what makes the position regime informative. Under Prop. 1 the naive gradient is identically zero there, and its correlation with the oracle is 0. The reported faithfulness is therefore 0.00 and not a 0.5 floor. The two committed estimators differ only in how the raw correlation is summarized across the six core games. Phase B reports the raw correlation directly (e.g. occlusion at $F = 0.639$), so a negative correlation is recorded as a negative number. In Phase A's single-unit and tuning analyses the per-game correlations can be negative, so we clip each per-game correlation at 0 and then average across games (e.g. the tuning analysis A3, lowest in Phase A at 0.091, the clip-at-zero mean of six per-game correlations that include negatives). For the same attribution map we also report, as quality measures, deletion/insertion area-under-curve and precision at k against the true causal top-k, $\mathrm{P@}k = |\,\mathrm{top}_k(\hat{E}) \cap \mathrm{top}_k(\{\Delta y(u)\})\,|/k$. For a graph-valued explanation (Phase A connectomics, Phase C circuit discovery) we use the edge F_1 against the true data-flow graph, the harmonic mean of precision and recall over recovered edges. For Phase C patching and mediation we use the causal-effect-agreement correlation against the true $\{\Delta y(u)\}$. The final $\mathrm{F}_{\text{score}}$ is the per-record primary metric averaged first within a game over its output records, then with equal weight across the six core games, so no game-rich method dominates; confidence intervals are 95% bootstrap over games (Section 3.5). The three Phases differ only in which primitive is primary: Phase A uses graph F_1 for the data-flow methods and the clip-at-zero mean correlation for the single-unit and tuning analyses, Phase B uses the raw effect-agreement correlation ρ with deletion/insertion AUC and $\mathrm{P@}k$ as the attribution map's quality measures, and Phase C uses the causal-effect-agreement correlation for patching and mediation and graph F_1 for circuit discovery. The leaderboard score is therefore a single scalar per method, but it is multi-metric in origin, and the per-method choice is recorded in the Supplement.

Sufficiency asks whether the explanation could regenerate the behavior under interventions it has not seen. We treat $\mathrm{S}_{\text{score}}$ as a held-out predictive estimator: $\hat{E}$ is fit (or its causal sites read off) on a calibration set of interventions and then asked to predict the true output y' under a disjoint test set of held-out $\mathrm{do}(u)$ edits, with the prediction compared to the bit-exact re-run. The score is the fraction of held-out interventions whose predicted y' matches the true y' within a fixed output-type tolerance (exact for discrete indices and events, a small ε band for continuous scores), so $\mathrm{S}_{\text{score}} \in [0, 1]$ and a value near zero means the explanation predicts no better than the unperturbed output. We deliberately do not define it through interchange interventions or scrubbing. Those are themselves mechanistic-interpretability methods under test in Phase C, so using them would make the criterion circular. Only Phase B and Phase C report S, and its bootstrap intervals follow Section 3.5.

Minimality asks whether the explanation is parsimonious and lives at the algorithmic level of the machine's own module decomposition. We define $\mathrm{M}_{\mathrm{score}} = |U^\star|/|U_{\hat{E}}| \in (0, 1]$, the ratio of the size of the true minimal cause set $U^\star$, the smallest set of registers, RAM cells, opcodes, or program variables whose intervention reproduces y, to the size $|U_{\hat{E}}|$ of the cause set the explanation actually names. An explanation that names exactly the minimal set scores 1; one that records the whole machine state pays for every spurious cell. The target level is the algorithmic unit, the register, RAM cell, opcode, module, or program variable, not the pixel. Whole-state recording names all 128 RAM cells, where the true cause set averages a handful. It therefore scores $\mathrm{M}_{\mathrm{score}} = 0.07$ across the six core games. Its faithfulness is perfect because it withholds nothing, but it is maximally non-minimal.

Two reporting axes carry the headline comparison, faithfulness against the ground truth and a plausibility proxy. The plausibility proxy is a documented, tradition-level value, not a human-subjects measurement, and we report it as such. It is assigned per method tradition, not per method instance, from the published evidence on how convincing each tradition's explanations are taken to be, before any faithfulness score on our machine is computed, so the proxy cannot be tuned to the result. The per-tradition assignment, its rubric, and its sources are tabulated in the Supplement (Section S7). There a sensitivity analysis perturbs the proxy under additive, rank-preserving, rank-jittering, and leave-one-tradition-out schemes. It finds the faithfulness-versus-proxy rank correlation stays negative throughout (from -0.79 to -0.40), so the headline contrast survives every alternative proxy assignment we tried. The proxy lets us find explanations with high plausibility but low faithfulness: explanations that convince but are wrong. F is satisfied by the wiring; S and M must be satisfied by the *computation*. Separating them lets us measure, later, what faithfulness alone leaves out.

3.5 Methods under test, games, and aggregation

We apply three traditions to the machine and score each by the triad of Eq. (3). The neuroscience battery (Phase A) runs connectomics and data-flow recovery, single-unit lesions, tuning curves, pairwise and global correlation structure, pooled-activity spectra, Granger causality, dimensionality reduction, and whole-state recording, scoring each against the known mechanism to quantify Kording's lesson at the low-faithfulness baseline [4]. The attribution and XAI battery (Phase B) runs gradient saliency and its variants, integrated gradients, occlusion, perturbation masks, randomized masking, LIME, and Shapley-style attribution [5, 6, 11–15]; Grad-CAM and its variants and attention rollout [9, 10, 41] require neural layers or a learned policy and are recorded as not applicable. The mechanistic-interpretability battery (Phase C) runs activation patching and causal mediation, interchange interventions and distributed alignment search, attribution and path patching, automatic circuit discovery, sparse autoencoders, causal scrubbing, linear probing with control tasks, and the logit and tuned lens [17–27, 43].

All faithfulness scoring runs on the six core games of Section 3.3, where the oracle's re-runs are exact. Cell- and pixel-level scores, correlation, deletion and insertion area-under-curve, and precision at k, are taken against T1 and need no T3; only object- and concept-level scores draw on the external T3 labels, and we flag every such score as alignment-to-given, not discovery. We aggregate the committed per-game records into one row per method on the shared faithfulness-versus-plausibility axes of the Results leaderboard. The headline contrast is the all-regime gap. Pooled over every output, causal and intervention methods average a faithfulness of 0.66 against 0.37 for gradient and correlational methods, a gap of 0.30 whose 95% interval excludes zero. On the position regime alone, where the naive gradient is zero under the hard forward map by Prop. 1, the two families average 0.49 against 0.25; this gap points the same way but is not significant at six games. The per-method extreme places an exact causal method, activation patching, at faithfulness 1.000 on all six games and the field's default saliency at only 0.165 on the position regime, the popular method nonetheless carrying the higher plausibility proxy (0.9 versus 0.5). We measure the semantic gap relative to these numbers.

Data availability. The interpretability scores reported here come from the committed analysis records in the repository. These records hold the per-method, per-game §R results and the aggregated cross-tradition leaderboard [35, 44]. Every quantitative claim in the Article reads from one of these committed artifacts. No number is re-run or estimated. The game cartridges used as the subject are third-party Atari 2600 ROMs that we do *not* redistribute. We follow the practice of the underlying emulator port [30]. We acquire the ROMs through the `AutoROM` retrieval procedure and report a SHA-256 hash for each ROM in the Supplementary Information. This lets others reproduce and verify the exact binaries independently. The recorded state trajectories used to train the sparse autoencoders and to run the patching and probing experiments are regenerated deterministically from those ROMs and the published action streams.

Code availability. The interpretability benchmark introduced here is available for review at https://github.com/akmaier/UnderstandingVCS. It comprises the scoring suite, the $\mathrm{F} \wedge \mathrm{S} \wedge \mathrm{M}$ triad metrics, the intervention oracle, and the per-phase method implementations. The repository also holds the reproducibility appendix that maps every reported number to its committed record and run command. A versioned snapshot will be archived on Zenodo with a minted DOI. An optional Code Ocean capsule reproduces the leaderboard and the figures from the committed result records alone. The benchmark rests on a differentiable, bit-exact emulator. It has two interchangeable versions (`jutari` in Julia, `jaxtari` in JAX/Python), and both are already public. The emulator is described in the companion port paper [30] (arXiv:2606.22447), and its source is openly available at the same repository. The companion paper supplies the bit-exactness validation and the gradient caveat that the present work imports. What is new here is the interpretability benchmark, the oracle scoring, and the semantic-gap diagnosis. We release the code under a permissive open-source licence so that a third party can score a new method against the same ground-truth oracle.

Author contributions. A.M. conceived the study, designed the ground-truth audit and the $\mathrm{F} \wedge \mathrm{S} \wedge \mathrm{M}$ triad, implemented the benchmark and oracle scoring, and wrote the manuscript. S.B. contributed to the method design and the analysis and revised the manuscript. P.K. contributed to the differentiable substrate and oracle and to the gradient-faithfulness analysis, and helped develop the differentiability framing. All

authors discussed the results, interpreted the semantic-gap finding, and approved the final manuscript.

Competing interests. The authors declare no competing interests.

Ethics and broader impact. This study uses no human-subjects data; the plausibility axis is a documented plausibility proxy by method tradition rather than a measurement, and we report it as such. The subject is a fully deterministic, hand-coded program with no learned policy, so no agent is trained. Interpretability methods are dual-use: a tool that audits a system can also be used to probe it. Our contribution is a calibration that makes explanation claims more trustworthy, and we release it openly to that end. The game cartridges are not redistributed.

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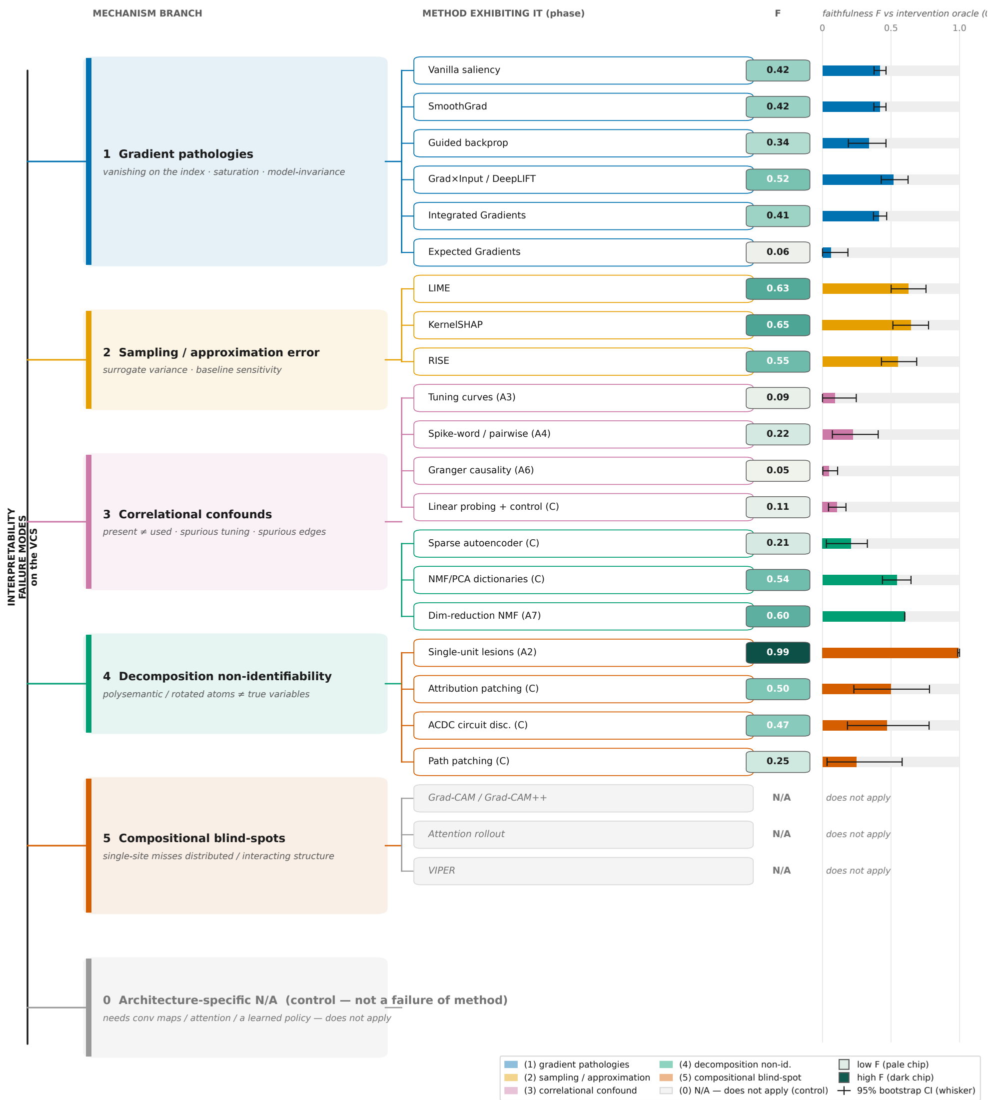

Fig. 4 A taxonomy of interpretability failure on the VCS, organised by mechanism. Each leaf is a method exhibiting the branch's failure mode, annotated with its *family-mean* faithfulness $F \pm 95\%$ bootstrap-over-games CI against the intervention oracle (Section 3.2; $0 =$ chance, $1 =$ oracle ceiling); the per-game exemplars quoted below carry their own "on ⟨game⟩" label. The lower branches confirm the expected failures: the naive gradient is provably zero on the discrete position/index regime (the family's position-regime mean is only 0.25, and stays low even after the sampler restores a gradient), and the architecture-specific methods (Grad-CAM, attention, VIPER) do not apply because the VCS lacks the neural substrate they require, a property of the subject, not a method failure. The higher-order branches are the *semantic gap*: present-but-unused concepts (linear probing [25] flags a present-but-unused cell the oracle says is inert), named-but-inert factorisations (the SAE [23] matches every variable at fraction 1.0 yet scores $F = 0.041$ on Pong), and graph-correct-but- incomplete circuits (ACDC [20] at $F = M = 1.0$, $S = 0.44$ on Breakout). The neural-network parallels shown alongside each branch are documented analogues [47–53], evidence transfer rather than proof. F is a point estimate and the whisker a 95% bootstrap CI over games; only the failure-exhibiting subset is drawn here, out of the full 31-row leaderboard (30 methods plus the oracle). Every annotated value reads from a committed record.

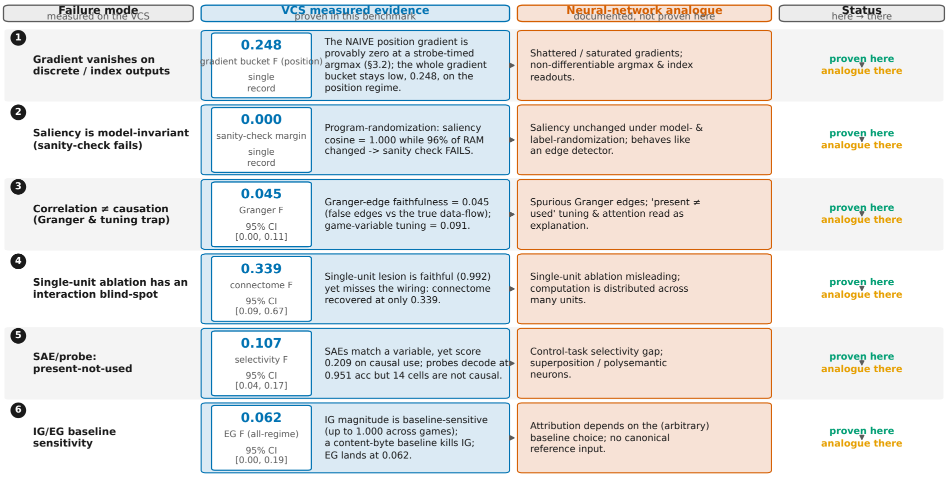

Fig. 5 **The VCS↔neural-network failure-mode map.** Each row pairs a failure mode we measured on the VCS against the oracle (Section 3.2) — left, blue — with the neural-network behavior it is a documented analogue of — right, vermillion. On the VCS the failure is provable because we hold the exact ground truth; the neural-network column is a documented analogue, not proven by this benchmark, and reads the attribution and probing literature ([3, 6, 47, 52] on gradients; [25, 48, 49, 53] on use vs. presence; [32, 50, 51] on probing, superposition, and reliability). The takeaway is one line: every VCS failure has a named counterpart in how interpretability methods are reported to fail on networks, so a property provable here is one to suspect there. On the position regime the causal and intervention methods open a 0.238 faithfulness gap over the gradient and correlational methods, in the expected direction but not significant at six games; the robust contrast is the all-regime gap of 0.30 whose interval excludes zero. The naive gradient is provably zero on the position read, and even after the sampler restores a gradient the position-regime mean stays low (about 0.25 for the gradient family); activation patching, by contrast, reaches the oracle ceiling (1.000) there where vanilla saliency reaches only 0.165. Per-row VCS scores, their committed records, and the full map are in the Supplementary Information.

Supplementary Information

This Supplementary Information records what the main paper summarizes. We give the per-method protocols (Section S1) and the leaderboard schema with its glossary (Section S2). We give the coverage and applicability of every method across games and output regimes (Section S3). We map each headline claim to its evidence (Section S4). We trace the provenance of every headline number down to the record file and command that produced it (Section S5). We state plainly what the paper does not claim (Section S6). We give the plausibility-proxy rubric and its sensitivity analysis (Section S7). The correctness triad $F \wedge S \wedge M$ used throughout is the one defined in the main Methods (Section 3.4); we reference it here rather than restating it. The benchmark scores 30 interpretability methods together with the intervention oracle (Section 3.2) as a positive control, for 31 leaderboard rows in total.

S1 Per-method protocols

This section records five things for each method under test. It records the exact objects the method consumes and produces, the reference it is scored against, its sampling or optimization budget, the metric that turns its output into a faithfulness score, and the committed record file that holds the per-game results. All methods share one evaluation contract. We state that contract once and then specialize it per method. Every method receives the same bit-exact VCS state on one of the six core games (`breakout`, `ms_pacman`, `pong`, `qbert`, `seaquest`, `space_invaders`). That state is not the boot screen. It is a shared gameplay state, produced by a single seeded random-action stream (seed 0) run for a 90-frame prefix and then rolled forward for a 15-frame analysis horizon. The stream is deterministic, so every method sees the identical state on each game, and the run is asserted bit-exact per game. An oracle cause-density gate screens the state before any method runs. The gate counts how many candidate causes move the output above a floor and rejects states with too few, so a near-flat oracle column cannot pass. This gate is what turns a degenerate frame into a usable one; on `seaquest` it raises the count of live candidate causes from 2 to 19. The candidate-cause set is the machine's own state. This set is the RAM bytes together with the CPU registers and the TIA and RIOT registers exposed by the substrate. The output every method attributes over is a shared screen-buffer region read at the end of the analysis horizon, not a single hand-picked pixel. The reference is the intervention oracle of Section 3.2, evaluated by exact re-runs of the emulator under $\mathrm{do}(u)$. A method that cannot be applied to a given output regime is recorded as not-applicable rather than scored as zero. An empty result is therefore never read as a wrong one. Faithfulness is the F axis of the main-text triad (Section 3.4); we reference that definition and do not restate it. Records live under `tools/xai_study/<phase>/out/`; the aggregate they feed is `tools/xai_study/compare/out/leaderboard.json`.

The neuroscience battery (Phase A) treats the running machine as a brain. It asks how far each classical analysis recovers the known mechanism on the shared gameplay state. Each method's baseline is the exact data-flow graph or causal role that the substrate makes available. Its budget is the six-game core unless noted, and its scoring metric is named in Table S1. One cell is honestly degenerate. On `ms_pacman` the

Table S1 Phase A protocols (neuroscience battery). One row per method. The columns give the input it reads, the reference it is scored against, the sampling or intervention budget, the scoring metric, and the committed record. All eight run on the six core games; faithfulness is the F axis of Section 3.4. The whole-state recording (A8) is included as a descriptive upper bound on coverage. Its faithfulness F is trivial, so we read it for its minimality M.

Method	Input / candidate causes	Reference	Scoring metric	Record
A1 connectomics	RAM/register activity over trajectory	data-flow graph	data-flow graph F_1 vs. oracle	A1
A2 lesions	single-cell knockouts via do(u)	true causal role	lesion-importance ρ vs. role	A2
A3 tuning curves	per-cell response vs. variable	variable coupling	$1-$ spurious-tuning rate	A3
A4 spike-word / pairwise	pairwise cell correlations	true coupling	corr. ρ vs. coupling	A4
A5 field potentials	pooled-activity spectra	true causal share	$1-$ clock-explained var.	A5
A6 Granger	directed lag structure	true data-flow	$1-$ false-edge rate	A6
A7 dim. reduction	NMF/PCA over activity	known variables	NMF matched-component frac.	A7
A8 whole-state	full state record	oracle decomp.	minimality M vs. oracle	A8

Records: `tools/xai_study/phaseA_kording/out/A{1..8}_*`; all eight run on the six core games.

pairwise analysis (A4) finds no coupled candidate pair to score at that frame, so its `ms_pacman` value is uninformative rather than a failure; the other five games carry it.

The attribution battery (Phase B) runs the everyday XAI toolbox over the same gameplay state. These methods produce a real-valued saliency or attribution over the candidate-cause set. We score that output by Pearson correlation against the oracle attribution. The gradient family runs with the bilinear sub-pixel sampler turned on, so it is scored on a real position gradient and not on the naive one. The naive gradient of the discrete position output is provably zero (Prop. 1); the sampler restores a finite gradient, but the restored gradient is low and mixed in faithfulness rather than a fix. Expected Gradients draws its reference pool from the states of the same seeded gameplay trajectory, not from a synthetic baseline. Two cells are honestly degenerate and recorded as such. The on-distribution counterfactual returns a zero delta on `space_invaders`, `ms_pacman`, and `qbert`, where its perturbation does not move the shared output at that frame; those cells are marked not-valid rather than scored. Expected Gradients scores low on the content regime of every game except `pong`, because the causal byte it should land on is static there. The budgets that matter are the Monte-Carlo or perturbation counts of the sampling estimators, recorded per method in Table S2. The seed-dispersion of the stochastic members is carried in `leaderboard_ci.csv` (`seed_sd`, `seed_sd_source`) and reproduced in Section S5.

The mechanistic battery (Phase C) runs the methods built to recover circuits. Each is scored against the true data-flow or the exact recovered-minus-exact deviation. The

Table S2 Phase B protocols (attribution battery). Each method emits a saliency over the candidate-cause set. We score it by Pearson correlation against the oracle attribution (Section 3.2). The reference and the six-game budget are shared. The budget column records the sampling or perturbation count that drives each estimator. The seed column flags which methods carry an independent-seed dispersion in `leaderboard_ci.csv`.

Method	Tradition	Budget / hyperparameters	Seed sd
Vanilla saliency	gradient	single backward pass	—
smoothgrad	gradient	noise-averaged gradient; σ-robustness committed	0.0
Guided backprop	gradient	single guided pass	—
Grad×Input / DeepLIFT	gradient	DeepLIFT reference-difference	—
Integrated Gradients	gradient	path integral, baseline = zero state	—
Expected Gradients	gradient	expectation over baselines (refits)	yes
kernelshap	gradient	Monte-Carlo coalition sampling ($\propto 1/\sqrt{N}$)	0.017
lime	gradient	local linear fit, perturbation refits	0.029
rise	gradient	random-mask Monte-Carlo ($\propto 1/\sqrt{N}$)	0.051
occlusion	intervention	exhaustive single-cell masking	—
Extremal perturbation	intervention	optimized mask, area-constrained	—
On-distribution counterfactual	intervention	on-manifold do(u) resampling	—

Records: `tools/xai_study/phaseB_attribution/out/`; all twelve run on the six core games and are scored by Pearson correlation against the oracle (Section 3.2). Seed sd is the independent-seed dispersion from `leaderboard_ci.csv`.

causal members of this phase are the ones that reach the oracle ceiling. We list their exact scoring metric so that the ceiling is not read as a tautology. Each member is scored against the substrate's known routine, and not against another method. Two members (NMF/PCA dictionaries and the sparse autoencoder) are dimensionality-reduction methods carried here for comparison. The sparse autoencoder has two aggregations that differ, which we reconcile in Section S5.

The intervention oracle is the positive control, not a method under test. It is the exact do(u) ground truth of Section 3.2, evaluated by re-running the bit-exact machine. It attains $F = S = M = 1.0$ by construction. Its record triple is the intervention oracle, the content-path gradient oracle, and their cross-check (`tools/xai_study/ground_truth/out/`). It anchors the ceiling against which every other row is read, which is why it occupies the thirty-first leaderboard row.

S2 Benchmark schema and glossary

The leaderboard is built from a flat record schema. Stating it removes the ambiguity about what a single number on the benchmark counts. Each method is run per game and writes a per-game record under its phase's `out/` tree. The leaderboard aggregates

Table S3 Phase C protocols (mechanistic battery). Each method is scored against the substrate's known circuit or the exact recovered-minus-exact deviation (Section 3.2), and not against another method. This is what keeps the oracle ceiling non-circular. The probing and dictionary-learning members are included for comparison with the causal members.

Method	Tradition	Scoring metric
Activation patching	causal	max $\lvert$recovered $-$ exact$\rvert$ vs. oracle
Causal scrubbing	causal	scrubbing-preserved performance
Interchange interventions (DAS)	causal	aligned interchange accuracy
Logit / tuned lens	causal	logit-lens fidelity to true intermediate
Path patching	causal	path-circuit F_1 vs. true routine
ACDC	causal	discovered-circuit best F_1 vs. dataflow
Attribution patching	gradient	corr. of approximation vs. exact patch
Linear probing (+ control)	probing	selectivity vs. control task
NMF / PCA dictionaries	dim. red.	matched-component fraction vs. vars
Sparse autoencoder	dim. red.	feature-variable matched fraction

Records: `tools/xai_study/phaseC_mechanistic/out/`; all ten run on the six core games.

those records into one row per method by averaging over games. The schema columns of a leaderboard row are given in Table S4. The aggregation rule is a mean over the n_{games} games. When a method emits several output records on one game, those records are first reduced within that game. The faithfulness, sufficiency, and minimality columns report the F, S, and M axes defined in Section 3.4. We do not re-derive those definitions here.

We fix the vocabulary that the schema implies, because the review asked for a single unambiguous method count. A *method* is one interpretability technique, e.g. `vanilla_saliency`. A *method row* is its single aggregated line on the leaderboard. A *per-game record* is one method scored on one game, written under a phase `out/` tree. An *output record* is a per-game record for one output regime; a method may emit several of these on one game. A *family mean* is the average over the members of a tradition family, e.g. causal/intervention. The *oracle positive control* is the exact do(u) ground truth of Section 3.2, scored as a method to anchor the ceiling. A *not-applicable method* is one that does not apply to a given output regime; it is recorded as N/A rather than as a zero. With this vocabulary the count is unambiguous. The benchmark scores 30 interpretability methods together with the intervention oracle as a positive control, for 31 leaderboard rows in total (`leaderboard.json.n_methods` = 31), aggregating 257 per-game records (`n_per_game_records_aggregated` = 257): 8 methods in Phase A, 12 in Phase B, 10 in Phase C, and the one oracle row. This count is the canonical one used in the abstract, the figures, and throughout this Supplement.

Table S4 Leaderboard row schema. The columns of one row in `compare/out/leaderboard.json` (`rows[]`) and its uncertainty companion `leaderboard_ci.csv`. The metric column names the per-method scoring function from Section S1; F, S, and M are the triad axes of Section 3.4. Aggregation is a mean over n_{games}, with bootstrap confidence intervals over games (and, where present, over records).

Column	Meaning
`phase`	phaseA_kording, phaseB_attribution, phaseC_mechanistic, or ground_truth
`method`	method identifier (the row key)
`tradition`	intervention, causal, gradient, correlational, `dim_reduction`, probing, descriptive, oracle
`n_games`	number of games the method was scored on (6 for the core set; 1 for the oracle)
`n_records`	number of per-game output records aggregated into the row
`faithfulness` (F)	faithfulness vs. the intervention oracle (Section 3.2), $[0, 1]$, higher better
`faithfulness_ci95`	95% bootstrap CI half-width over games
`faithfulness_position_regime`	F restricted to the discrete sprite-position/index regime
`faithfulness_content_regime`	F restricted to the continuous content regime
`triad_S` / `triad_M`	sufficiency / minimality axes (Section 3.4), where applicable
`plausibility_proxy`	documented method-tradition proxy, $[0, 1]$ — not a human-subjects measurement
`is_positive_control`	1 for the oracle row, 0 otherwise
`metric_names`	the per-method scoring function (Section S1)
`commits`	the commit(s) at which the per-game records were produced

S3 Coverage and applicability

The benchmark scores methods on a six-game core drawn from a larger bit-exact substrate. The coverage table makes the scope of that choice explicit. Paper 1 validates the differentiable VCS ports bit-for-bit against the `xitari` reference on all 64 Arcade Learning Environment games, within a 30-frame RAM and 60-frame screen window. The interpretability study reported here runs on the six games listed in Table S5. Every method is scored on a shared gameplay state, produced by one seeded random-action stream (seed 0, 90-frame prefix, 15-frame analysis horizon), not on the boot screen. This keeps the analysis inside the Paper 1 bit-exact window while giving the oracle a rich, non-degenerate cause set to score against. We pick these six games because all three tiers of ground truth (Section 3.3) are available for them and the oracle is exact. The remaining 58 games are bit-exact substrate but are not scored here. Some fall outside the validated horizon for the analyses we run. For others, the tiered labels they would need are not yet produced. They are candidates for the larger sweep, and are not excluded for any methodological reason. The horizon column records the validated window. The tier column records which of the three ground-truth tiers (T1 exact intervention, T2 data-flow, T3 externally-supplied semantic labels) is available.

A gameplay state can still be degenerate for a particular method, and the benchmark records this rather than scoring it as a failure. The oracle cause-density gate is the first guard. It counts how many candidate causes move the shared output above a floor, and rejects a state that does not clear a minimum count, so a near-flat oracle column never reaches a method. On `seaquest` the gate raises the count of live candidate causes at the analysis frame from 2 to 19, which is what makes the game usable.

Table S5 Core-game coverage. The six core games carry exact tiered ground truth and the full method battery. Faithfulness is scored within the Paper 1 bit-exact window. The last two columns give the methods run and the per-game output records contributed. The wider 64-game substrate is bit-exact (Paper 1) but is reserved for the larger sweep. T3 labels are external (Section 3.3); we record this so the benchmark does not appear to certify them.

Game	Paper-1 status	Horizon	Tiers	Methods run	Records
breakout	bit-exact	30/60 fr	T1,T2,T3	30 + oracle	per-game
ms_pacman	bit-exact	30/60 fr	T1,T2,T3	30 + oracle	per-game
pong	bit-exact	30/60 fr	T1,T2,T3	30 + oracle	per-game
qbert	bit-exact	30/60 fr	T1,T2,T3	30 + oracle	per-game
seaquest	bit-exact	30/60 fr	T1,T2,T3	30 + oracle	per-game
space_invaders	bit-exact	30/60 fr	T1,T2,T3	30 + oracle	per-game
58 further games	bit-exact (P1)	30/60 fr	T1,T2 partial	reserved	—

Three residual degenerate cells survive the gate and are marked, not scored as zero. On `ms_pacman` the pairwise neuroscience analysis (A4) finds no coupled candidate pair at that frame, so its cell is uninformative. The on-distribution counterfactual returns a zero delta on `space_invaders`, `ms_pacman`, and `qbert`, where its perturbation does not move the shared output; those cells are recorded as not-valid. Expected Gradients scores low on the content regime of every game but `pong`, because the causal byte it should attribute to is static there. None of these cells is read as a wrong answer.

Whether a method applies at all depends on what the method needs from its target. The applicability matrix records that for every method family. The Phase A neuroscience analyses read the raw machine state and need neither a learned policy nor gradients. The Phase B attribution methods split into two groups. The gradient members need a differentiable forward pass. The intervention members need only the ability to set state. The Phase C mechanistic methods need access to internal activations. The causal members among them also need the ability to intervene. Table S6 records, per family, whether the method operates on the raw VCS, whether it requires a learned policy, gradients, internal layers or attention, or external labels, whether it uses oracle-like interventions, and whether it receives a supplied hypothesis. The table shows the same pattern the results confirm. The families that score high on faithfulness are the ones that can intervene. The families that collapse on the position regime are the gradient and correlational ones.

S3.1 T3 semantic-label coverage across the 64-game roster

The tier-3 (T3) ground truth is externally supplied: it consists of semantic object and position labels imported from AtariARI [32] and OCAtari [33], not derived by our oracle. This subsection documents where such labels *exist*, so that the breadth and position analyses can, in principle, scale beyond the six-game core of Table S5. It reports label *availability and provenance* only. It does not report any result on the wider battery. The larger sweep is in progress and has produced no scores yet; nothing below should be read as a score.

Table S6 Per-family applicability. What each method family requires of its target. "Intervenes" marks oracle-like do(u) access; "hypothesis" marks methods that receive a supplied circuit hypothesis to test. The raw-VCS column distinguishes analyses that read machine state directly from those that need a differentiable or layered model. A blank means the requirement does not apply.

Family	raw VCS	policy	gradients	layers/attn	intervenes	hypothesis
A neuroscience	yes	no	no	no	A2 only	no
B gradient	yes	no	yes	no	no	no
B intervention	yes	no	no	no	yes	no
C causal	yes	no	no	yes	yes	yes
C gradient (attr. patch)	yes	no	yes	yes	approx.	yes
C probing	yes	no	no	yes	no	no
C / A dim. reduction	yes	no	no	yes	no	no
oracle (control)	yes	no	no	yes	exact	—

Table S7 T3 semantic-label coverage. Availability of external position/object labels across the 64-game bit-exact roster, computed against the installed OCAtari object tables and the AtariARI `atari_dict`. This is a provenance record of which games *could* enter a label-dependent analysis; it reports no result on any of them. Every AtariARI-labeled game is also OCAtari-labeled, so OCAtari alone defines the maximal labeled set of 54.

Set	Count	Note
Our bit-exact roster	64	all Paper-1 games
OCAtari-labeled (offset-corrected)	54	largest labeled subset
AtariARI-labeled	22	all a subset of the OCAtari 54
Covered by both sources	22	richest (cross-checkable)
AtariARI-only	0	every AtariARI game is in OCAtari
No T3 label at all	10	excluded from label-dependent scoring

The counts are computed, not estimated. We intersect our bit-exact roster with the games that carry a label in the installed source of each project: the OCAtari object-property tables and the AtariARI `atari_dict`. OCAtari is preferred for render-aligned position labels because its RAM mode applies the render-time offsets that correct the raw-RAM misalignment [33]; AtariARI labels are cross-checkable but do not carry those offsets. Table S7 summarizes the intersection. Of the 64 bit-exact games, 54 carry an OCAtari label, which is the maximal labeled subset: every one of the 22 AtariARI-labeled games is also in the OCAtari 54, so the AtariARI-only count is 0 and OCAtari alone fixes the ceiling. The 22 games covered by both sources are the richest and the only cross-checkable ones. Ten games carry no T3 label from either source and are excluded from any label-dependent scoring. In percent of the roster, OCAtari labels 84%, both sources cover 34%, and 16% have no label at all.

For traceability we enumerate the sets. The 54 OCAtari-labeled games are: `air_raid`, `alien`, `amidar`, `assault`, `asterix`, `asteroids`, `atlantis`, `bank_heist`, `battle_zone`, `beam_rider`, `berzerk`, `bowling`, `boxing`, `breakout`, `carnival`, `centipede`, `chopper_command`, `crazy_climber`, `demon_attack`, `double_dunk`, `enduro`, `fishing_derby`, `freeway`, `frostbite`, `gopher`, `hero`, `ice_hockey`, `jamesbond`, `kangaroo`, `krull`, `kung_fu_master`, `montezuma_revenge`, `ms_pacman`, `name_this_game`, `pacman`, `phoenix`, `pitfall`, `pong`, `pooyan`, `private_eye`, `qbert`, `riverraid`, road_runner, robotank, seaquest, skiing, space_invaders, star_gunner, tennis, time_pilot, up_n_down, venture, video_pinball, and yars_revenge. The 22 games covered by both sources are: asteroids, berzerk, bowling, boxing, breakout, demon_attack, freeway, frostbite, hero, montezuma_revenge, ms_pacman, pitfall, pong, private_eye, qbert, riverraid, seaquest, space_invaders, tennis, venture, video_pinball, and yars_revenge. The 10 games with no T3 label, dropped from any label-dependent scoring, are: defender, elevator_action, gravitar, journey_escape, solaris, surround, tutankham, videochess, wizard_of_wor, and zaxxon.

S4 Claims and evidence

Every headline claim in the paper rests on a specific artifact. The table in this section makes that dependency checkable. For each claim we list the figure or table that supports it, the games and methods it uses, the metric, the artifact it traces to, and the limitation that bounds it. Faithfulness, sufficiency, and minimality are the three axes of Section 3.4. The numbers themselves are sourced in Section S5. The table below names which evidence supports which claim; it does not repeat the values. One split matters for how the table is read. The all-regime faithfulness gap between causal/intervention and gradient/correlational methods is 0.297 with a 95% interval of $[0.232, 0.375]$ that excludes zero, and it is the primary evidence for the headline. The position-only gap is 0.238 with an interval of $[-0.049, 0.315]$ that crosses zero at six games. We report the position gap as directional support, not as a significant result.

S5 Number provenance

Every headline number traces to a committed record, the script that produced it, and the command that re-derives it. This section is the in-paper copy of that traceability table. It mirrors Section 6.5 of the reproducibility document (xai_2_interpretability/REPRODUCIBILITY.md). The values are the measured, committed ones at the current revision. The confidence intervals are read from tools/xai_study/compare/out/leaderboard_ci.csv rather than recomputed here. Each record also carries seed, where, commit, and timestamp, so the provenance of every value is self-documenting.

One number needs a reconciliation, and we record it rather than hide it. The sparse autoencoder is the only method whose two aggregations disagree. The over-games mean is $F = 0.1568$ with a 95% bootstrap CI of [0.0282, 0.3274]. The over-records mean is 0.2088 with CI [0.0241, 0.4945]. Both are read from leaderboard_ci.csv. The figures and the main-text leaderboard plot the over-records value 0.2088, which lies inside the over-games interval. We treat the over-games mean as the canonical row value for the aggregation rule of Section S2, and report the over-records figure alongside it, so that text, table, and figure agree on which aggregation each cites. No headline claim depends on which of the two is read, since the sparse autoencoder sits near the floor under both.

Table S8 Claims and their evidence. Each headline claim maps to the figure or table that supports it, the games and methods involved, the metric, the committed artifact, and the limitation that bounds the claim. The artifact column points to the record that Section S5 traces to a value. The limitation column states the scope, so the claim is not read wider than the evidence.

Claim	Evidence	Methods / games	Metric	Limitation
Causal/intervention methods beat gradient/correlational on faithfulness across all regimes	Fig. 2; leaderboard	25 methods, 6 games	F mean (gap 0.297)	robust: gap 0.297 $[0.232, 0.375]$ excludes zero; this all-regime gap is the primary evidence
The gap holds with the same sign on the discrete position regime	Fig. 2; `faithful_demo`	13 rows, 6 games	F position (gap 0.238)	directional only: gap 0.238 $[-0.049, 0.315]$ crosses zero at 6 games, so it is not significant
A faithful causal method reaches the oracle ceiling where the default tool stays low	`faithful_demo`; Fig. 2	act. patching vs. vanilla saliency, 6 games	F 1.0 vs. 0.165 (position regime; all-regimes saliency F 0.419, plotted in Fig. 2)	one decisive pair
The naive gradient of a discrete position output is provably zero	Prop. 1 (Section 3.2); `oracle_grad`	all gradient methods	grad@position = 0; sampler restores 166	position path only; restored gradient is low/mixed
High plausibility coexists with low faithfulness (the danger zone)	Fig. 2; `faithful_demo`	vanilla saliency	proxy 0.9, F 0.165 (position regime; all-regimes F 0.419, plotted in Fig. 2)	proxy, not measured plausibility
Faithful attribution recovers the wiring but not the computation's semantics	Fig. 4 (Section S3); discussion	Phase B vs. C	F vs. S, M	necessary-condition stress test
The failure modes have documented NN analogues	Fig. 5	taxonomy	—	analogues, not proof

S6 What this paper does not claim

The benchmark is sharp because its scope is narrow. Stating the boundaries keeps the result from being read wider than the evidence allows. We do not claim to recover a program's semantics from scratch. The study measures whether an explanation names the true causes and regenerates the behavior on a machine whose answer is already known. It does not show that any method discovers the meaning of a routine without that ground truth in hand. A faithful method recovers the causal structure but not the meaning of the variables, and this remaining gap is the point rather than a result to be explained away.

We do not evaluate learned agents. The subject throughout is the bit-exact VCS itself. The candidate causes are its registers, RAM cells, and program variables, not the weights of a trained policy. The neural-network parallels drawn in the discussion and in Fig. 5 are documented analogues from the literature, not claims proven on this benchmark. We label them as such, so that a reader does not take a mapped failure mode for a demonstrated one.

We do not prove that neural-network interpretability is impossible. The VCS gap is a necessary-condition stress test. A method that fails to recover the causal structure of a fully known machine cannot be trusted to recover it on an unknown one. Yet, a method that succeeds here is not thereby certified on networks. The "floor" framing is an evidence-transfer argument, not a proven lower bound on neural-network interpretability.

We do not measure human plausibility. The plausibility axis is a documented method-tradition proxy, computed from a fixed rubric rather than collected from raters, and we report it as a proxy throughout. Its job is to expose the danger zone where a convincing explanation is unfaithful, not to stand in for a human-subjects study. We also do not redistribute the ROMs. The substrate is validated against an external reference, and the records are reproducible from committed artifacts without the ROMs. The ROM provenance is given as hashes, so a third party supplies their own legally obtained copies.

S7 Plausibility-proxy rubric and its sensitivity

The plausibility axis of the leaderboard is a documented, tradition-level proxy, not a human-subjects measurement. This section records how each value was assigned. It also shows that the headline contrast does not depend on the particular numbers chosen. The Methods (Section 3.4) name two reporting axes. Faithfulness is measured on the machine against the intervention oracle of Section 3.2. The plausibility proxy stands in for how convincing a tradition's explanations are usually taken to be. We assign the proxy once per method tradition, before any faithfulness score on our machine is computed, so it cannot be tuned to the result. Its only job is to expose the danger zone of explanations that look convincing yet are unfaithful. For that job it must be coarse, fixed in advance, and sourced rather than precise.

The rubric assigns one proxy per tradition on a fixed five-step scale. It scores two documented properties of each tradition's typical output and takes their average. The first property is whether the explanation produces a dense, picture-like artifact that a reader sees as an account of the computation. Heatmaps and saliency-style maps over the input are read as convincing by construction. This is the perceptual-immediacy axis. The second property is whether the field has historically treated the tradition's output as a self-evident explanation rather than a hypothesis to be tested. This is the adoption-as-explanation axis. A tradition that scores high on both, such as gradient saliency, receives a high proxy. A tradition whose output is an austere intervention table or a circuit edge list, rarely mistaken for a finished story, receives a low one. The scale is deliberately blunt, $\{0.3, 0.5, 0.55, 0.75, 0.8, 0.85, 0.9\}$. A proxy meant to expose a danger zone gains nothing from false precision, and a finer scale would invite over-reading. Every value in Table S10 traces to the `plausibility_proxy` field of the per-method rows in `compare/out/leaderboard.json`. The table only groups those rows by tradition and records the rationale.

The paper draws a contrast between this proxy and measured faithfulness. On the committed records the two move in opposite directions. Across the thirty methods under test the Spearman rank correlation between faithfulness and the proxy is -0.36, a negative association. The traditions the field finds most convincing are, on this machine, among the least faithful. Eight methods fall in the danger zone, defined here as a proxy of at least 0.70 together with a faithfulness below 0.40. They include the correlational neuroscience methods, Expected Gradients, guided backprop, linear probing, and the sparse autoencoder. The per-method extreme makes the point sharply. Vanilla saliency has the highest proxy, 0.9, yet scores faithfulness 0.165 on the position regime. Activation patching has a lower proxy, 0.5, and scores 1.0. Here the less faithful method is the more plausible one. These figures are produced by `tools/xai_study/compare/plausibility_sensitivity.py` from the leaderboard and recorded in `compare/out/plausibility_sensitivity.csv`.

A proxy fixed by a coarse rubric raises the question of whether the contrast is an artifact of the particular numbers. The answer is that it is not. We perturb the proxy under four deterministic schemes and recompute the relationship each time. The analysis uses fixed, index-keyed offsets rather than random draws, so it re-derives bit-for-bit. The first scheme adds a fixed per-method jitter of magnitude $\sigma \in \{0.05, 0.1, 0.2\}$, alternating in sign by row. The second compresses every proxy toward the common mean by a fraction σ, which preserves the rank order of traditions. The third uses the same index-keyed offsets at $\sigma \in \{0.1, 0.2\}$, large enough to swap neighbouring traditions' ranks. The fourth drops one whole tradition at a time and recomputes on the remainder. Across all fourteen perturbations the rank correlation between faithfulness and proxy stays negative. It ranges from -0.76 to -0.08. The danger zone stays populated, with between four and ten methods. The per-method anchor sign holds in every scheme where both anchor methods are present, the more plausible saliency remaining the less faithful. Two runs bound the range. The strongest is the leave-the-gradient-tradition-out run. It removes the most plausible and least faithful block at once, and still leaves six danger-zone methods and a rank correlation of -0.76. The weakest is the leave-the-causal-tradition-out run, where the correlation falls to -0.08, close to zero; removing the faithful causal block flattens the association, so the contrast is carried in part by that block. Table S11 reports the full sweep. The contrast keeps its sign under every alternative proxy assignment we tried: plausibility and faithfulness are not the same axis, and high plausibility is no guarantee of faithfulness.

Table S9 Headline-number provenance. Each number → its committed record and JSON key → the generating or aggregating script → the command that reproduces it. Values are the measured, committed numbers at this revision (consistent with REPRODUCIBILITY Section 6.5); CIs are read from `leaderboard_ci.csv`. Paths are relative to `tools/xai_study/`.

Number	Value	Record (JSON key)	Command
Faithfulness gap, all regimes (causal − gradient)	0.2971 $[0.232, 0.375]$, excl. 0	`leaderboard_ci.csv`: `all_regime_gap` mean	`uncertainty.py`
Causal/interv. mean F, all regimes	0.6644 $[0.599, 0.741]$, n=11	`leaderboard_ci.csv`: `family_causal_intervention` mean	`uncertainty.py`
Gradient/corr. mean F, all regimes	0.3673 $[0.324, 0.411]$, n=14	`leaderboard_ci.csv`: `family_gradient_correlational` mean	`uncertainty.py`
Faithfulness gap, position regime	0.2382 $[-0.049, 0.315]$, crosses 0	`leaderboard_ci.csv`: `position_regime_gap` mean	`uncertainty.py`
Causal/interv. mean F, position regime	0.4859 $[0.227, 0.560]$, n=4	`leaderboard_ci.csv`: `family_..._intervention_position` mean	`uncertainty.py`
Gradient/corr. mean F, position regime	0.2477 $[0.142, 0.386]$, n=9	`leaderboard_ci.csv`: `family_..._correlational_position` mean	`uncertainty.py`
Decisive pair, position regime (act. patch vs. vanilla sal.)	1.000 vs. 0.165	`faithful_demo`: `pair_contrast`	`faithful_demo.py`
Plausibility proxy, decisive pair	0.5 vs. 0.9	`faithful_demo`: `...plausibility_proxy`	`faithful_demo.py`
ACDC sufficiency S	0.3096	`leaderboard_ci.csv`: ACDC `S_mean`	`uncertainty.py`
Whole-state minimality M	0.0768	`leaderboard_ci.csv`: A8 `M_mean`	`uncertainty.py`
Methods scored (rows)	31	`leaderboard`: `n_methods`	`leaderboard.py`
Per-game records aggregated	257	`leaderboard`: `n_per_game_records_aggregated`	`leaderboard.py`
Naive gradient, discrete position output	position 0; content 1.0; sampler 166	`oracle_grad_pong.json`: `value`, `extra.sampler`	`oracle_grad.jl`
Exact oracle max $\lvert\Delta\text{score}\rvert$ (Pong)	17.0	`oracle_pong_score.json`: `value`	`oracle_intervene.jl`

Scripts live under `tools/xai_study/compare/` (Python) and `tools/xai_study/ground_truth/` (Julia); records under the corresponding `out/` trees. JSON keys are abbreviated with "`...`" where nested; the full paths are in REPRODUCIBILITY Section 6.5.

Table S10 Plausibility-proxy rubric. The proxy assigned to each method tradition, with the documented rationale. Values are the `plausibility_proxy` field of the leaderboard rows (`compare/out/leaderboard.json`). They are tradition-level and fixed before any faithfulness score is computed, so the proxy cannot be tuned to the measured faithfulness. A higher value means an explanation that the field tends to find more convincing, not one that is more faithful. The oracle (Section 3.2) is the positive control and is not a method under test.

Tradition	Proxy	Rationale (documented basis)
gradient	0.90	Dense input-space heatmaps read as a visual account of the decision; the most widely adopted XAI outputs, routinely shown as if self-explanatory.
probing	0.85	A trained decoder that "finds" a concept reads as evidence the concept is used; high adoption, though decodability is not causal use.
correlational	0.80	Tuning curves and coupling spectra match the neuroscience idiom of a convincing single-unit account.
dim. reduction	0.75	Low-dimensional factors and dictionary atoms look like a clean summary of the representation.
intervention	0.55	Occlusion and counterfactual outputs are quantitative tables, less immediately picture-like, treated as tests rather than stories.
causal	0.50	Circuit edge lists and patching results are austere artifacts rarely mistaken for a finished narrative.
descriptive	0.30	Whole-state recording is an explicit non-explanation: it withholds nothing and reads as a dump, not an account.
oracle (control)	1.00	The Section 3.2 ground truth by construction; positive control, not a method under test.

Table S11 Proxy-sensitivity sweep. The faithfulness-versus-proxy relationship under each perturbation, from `compare/out/plausibility_sensitivity.csv`. "ρ" is the Spearman rank correlation between faithfulness and proxy over the methods present. "danger" is the count of high-proxy (≥ 0.70), low-faithfulness (< 0.40) methods. "anchor" is 1 when the more plausible saliency is the less faithful of the saliency/activation-patching pair, and blank where a leave-one-out run removes an anchor. The baseline rubric is the first row. Across every scheme the correlation stays negative and the danger zone stays populated.

Scheme	Param	n	ρ	Danger	Anchor
baseline	—	30	−0.36	8	1
additive jitter	$\sigma = 0.05$	30	−0.40	8	1
additive jitter	$\sigma = 0.10$	30	−0.41	8	1
additive jitter	$\sigma = 0.20$	30	−0.42	10	1
rank-preserving	$\sigma = 0.10$	30	−0.36	8	1
rank-preserving	$\sigma = 0.20$	30	−0.36	8	1
rank-jittering	$\sigma = 0.10$	30	−0.41	8	1
rank-jittering	$\sigma = 0.20$	30	−0.42	10	1
leave-out causal	—	24	−0.08	8	
leave-out correlational	—	26	−0.40	4	1
leave-out descriptive	—	29	−0.29	8	1
leave-out dim. red.	—	27	−0.34	7	1
leave-out gradient	—	20	−0.76	6	
leave-out intervention	—	25	−0.34	8	1
leave-out probing	—	29	−0.36	7	1